



LEVEL-II (C.W)

FOCUS, VERTEX, DIRECTRIX, LATUSRECTUM, AXIS, FOCAL CHORD, FOCAL DISTANCE, DOUBLE ORDINATE

1. The ends of latusrectum of a parabola are $(-3,1)$ and $(1,1)$ then one of the equation of parabola is

1. $(x+1)^2 = 4y$ 2. $(x-1)^2 = 4y$

3. $(x+1)^2 = 2y$ 4. $(x-1)^2 = 2y$

2. The length of the latusrectum of the parabola

$$169[(x-1)^2 + (y-3)^2] = (5x-12y+17)^2 \text{ is}$$

1) $14/13$ 2) $7/13$ 3) $28/13$ 4) $56/13$

3. The length of double ordinate of parabola $y^2 = 8x$ which subtends an angle 60° at vertex is

1. $4\sqrt{3}$ 2. $8\sqrt{3}$ 3. $16\sqrt{3}$ 4. $32\sqrt{3}$

4. Tangents are drawn from $P(3,0)$ to the circle $x^2 + y^2 = 1$ touches the circle at points A and B. The equation of locus of the point whose distances from the point P and the line AB are equal is

1) $9y^2 + 48x - 80 = 0$ 2) $9y^2 - 48x + 80 = 0$

3) $3y^2 - 12x + 39 = 0$ 4) $3x^2 - 12y + 39 = 0$

5. The focus of parabolic mirror is at a distance of 5cm from its vertex if the mirror is 45cm deep, the distance AB

1) 30cm 2) 40cm 3) 50cm 4) 60cm

6. The length of the double ordinate of the parabola $y^2 - 8x + 6y + 1 = 0$ which is at a distance of 32 units from vertex is

1) 28 2) 30 3) 32 4) 26

7. If the equation

$$25\{(x-5)^2 + (y-3)^2\} = (3x-4y+1)^2$$

represents a parabola then its axis is

1) $4x+3y-10=0$ 2) $4x+3y-15=0$

3) $4x+3y-29=0$ 4) $4x+3y-17=0$

8. The value of p such that the vertex of $y = x^2 + 2px + 13$ is 4 units above the x-axis is

1) 2 2) ± 4 3) 5 4) ± 3

9. The vertex of a parabola is $(2, 0)$ and its directrix is y-axis. The end of latusrectum in the first quadrant is

1) $(2, 4)$ 2) $(4, 4)$ 3) $(6, 4)$ 4) $(8, 4)$

10. If two circles $x^2 + y^2 - 6x - 6y + 13 = 0$ and $x^2 + y^2 - 8y + 9 = 0$ intersect at A and B. The focus of the parabola whose directrix is line AB and vertex at $(0, 0)$ is

1) $\left(\frac{3}{5}, \frac{1}{5}\right)$ 2) $\left(-\frac{3}{5}, \frac{1}{5}\right)$

3) $\left(-\frac{3}{5}, -\frac{1}{5}\right)$ 4) $\left(\frac{3}{5}, -\frac{1}{5}\right)$

EQUATION OF PARABOLA/CONIC

11. The equation of the parabola with axis $3x + 4y - 4 = 0$ the tangent at the vertex $4x - 3y + 7 = 0$ and with length of latusrectum 4 is

1) $(3x + 4y - 4)^2 = 10(4x - 3y + 7)$

2) $(3x + 4y - 4)^2 = 4(4x - 3y + 7)$

3) $(3x + 4y - 4)^2 = 20(4x - 3y + 7)$

4) $(3x + 4y - 4)^2 = 5(4x - 3y + 7)$

12. Let $y = f(x)$ be a parabola, having its axis parallel to y-axis, which is touched by the line $y = x$ at $x = 1$, then

1) $2f(0) + f'(0) = 1$

2) $f(0) + f'(0) + f''(0) = 1$

3) $f'(1) = 2$ 4) $f'(0) = f'(1)$

13. The focus of a parabola is $(1, 2)$ and the point of intersection of the directrix and axis is $(2, 3)$. Then the equation of the parabola is

1) $(x-1)^2 + (y-2)^2 = \frac{1}{4}(x+y-5)^2$

2) $(x-1)^2 + (y-2)^2 = \frac{1}{2}(x+y-5)^2$

3) $(x-1)^2 + (y-2)^2 = \frac{1}{5}(x+y-5)^2$

4) $(x-1)^2 + (y-2)^2 = \frac{1}{25}(x+y-5)^2$

14. P is a point which moves in the xy plane such that the point P is nearer to the centre of a square than any of the sides. The four vertices of the square are $(\pm a, \pm a)$. The region in which P will move is bounded by parts of parabolas of which one has the equation

1) $y^2 = a^2 + 2ax$ 2) $x^2 = a^2 + 2ay$
3) $y^2 + 2ax = a^2$ 4) All of these

15. The number of parabolas passing through the three points (1, 3), (6, 13), (-5, -9) is

1) 3 2) 2 3) 0 4) infinite

16. The graph represented by the equation

$x = \sin^2 t, y = 2 \cos t$ is

1) a portion of a parabola 2) a parabola
3) a part of sine graph 4) a part of hyperbola

TANGENT AND ITS APPLICATIONS

17. The locus of the centroid of triangle formed by a tangent to $y^2 = 36x$ with coordinate axes is

1) $y^2 = -9x$ 2) $y^2 = -3x$
3) $y^2 = 3x$ 4) $y^2 = 9x$

18. If the line $7x + 6y - 13 = 0$ touches the parabola $y^2 - 7x - 8y + 14 = 0$ then the point of contact is

1) (2, 1) 2) (1, -1) 3) (-1, 1) 4) (1, 1)

19. The straight line $x + y = k + 1$ touches the parabola $y = x(1 - x)$ if

1) $k = -1$ 2) $k = 0$
3) $k = 1$ 4) k takes any real value

20. If two tangents drawn from the point (α, β) to the parabola $y^2 = 4x$ be such that the slope of one tangent is double of the other then

1) $\beta = \frac{2}{9}\alpha^2$ 2) $\alpha = \frac{2}{9}\beta^2$
3) $2\alpha = 9\beta^2$ 4) $\alpha = 2\beta^2$

21. Equation of the common tangent to $y^2 = 4x$ and $x^2 = 32y$ is

1) $x + 2y + 4 = 0$ 2) $x + 2y + 8 = 0$
3) $2x + y + 8 = 0$ 4) $2x + y - 16 = 0$

22. The locus of point of intersection of the two tangents to $y^2 = 4ax$ inclined at an angle 45° is

1) $x^2 + y^2 - 4ax + 2a^2 = 0$ 2) $x^2 - y^2 + 6ax + a^2 = 0$
3) $x^2 + y^2 + 8ax + 4a^2 = 0$ 4) $x^2 + y^2 - 2ax + 4a^2 = 0$

23. Area of the triangle formed by the pair of tangents drawn from (-1, 4) to $y^2 = 16x$ and the chord of contact of (-1, 4) is

1) $8\sqrt{2}$ 2) $16\sqrt{3}$
3) $5\sqrt{2}$ 4) $16\sqrt{2}$

24. Locus of the point of intersection of perpendicular tangents drawn one each to the parabolas $y^2 = 4(x+1), y^2 = 8(x+2)$ is

1) $x + 12 = 0$ 2) $x + 8 = 0$
3) $x + 4 = 0$ 4) $x + 3 = 0$

25. The locus of point of intersection of two tangents to $y^2 = 4ax$ at t and 2t on the parabola is

1) $2y^2 = 9ax$ 2) $4y^2 = 9ax$
3) $3y^2 = 4ax$ 4) $3y^2 = 8ax$

26. Two tangents to parabola $y^2 = 4ax$ have inclinations θ_1 and θ_2 with x-axis such that $\tan^2 \theta_1 + \tan^2 \theta_2 = k$ then the locus of the point of intersection is

1) $y = kx$ 2) $y^2 = kx^2 + 2ax$
3) $y^2 = 2ax$ 4) $y^2 = 2a$

27. Equation of the common tangent to the circle $x^2 + y^2 = 4ax$ and $y^2 = 4ax$ is

1) $x + y + a = 0$ 2) $x = 0$
3) $x = a$ 4) $x - y + a = 0$

28. Equation of the two tangents drawn from (1, 4) to the parabola $y^2 = 12x$ are

1) $x - y + 3 = 0, 3x - y + 1 = 0$
2) $x - y + 1 = 0, x - 2y + 4 = 0$
3) $x + y - 2 = 0, x - y = 3$
4) $x + y - 1 = 0, x + 2y + 4 = 0$

29. The angle between the two tangents drawn from origin to the parabola $y^2 = 4a(x - a)$ is

1) 90° 2) 30°
3) $\tan^{-1}(2)$ 4) 45°

30. Two tangents to $y^2 = 4ax$ make angles θ_1, θ_2 with x axis. If $\cos \theta_1 \cos \theta_2 = k$ then the locus of their intersection is

- 1) $x^2 = k^2[(x-a)^2 + y^2]$ 2) $x^2 = k^2[(x+a)^2 + y^2]$
 3) $x^2 = k^2[(x-a)^2 + 4y^2]$ 4) $4x^2 = k^2[(x+a)^2 + y^2]$

31. If a tangent is drawn to the parabola $y^2 = 4x$ through $(-2, 1)$ then the point of contact is

- 1) $(2, 1)$ 2) $(1, 2)$ 3) $(1, -2)$ 4) $(2, \sqrt{2})$

32. The equation of a tangent to the parabola $y^2 = 8x$ is $y = x + 2$, The point on this line from which the other tangent to the parabola is perpendicular to the given tangent is

- 1) $(-2, 0)$ 2) $(-1, 1)$ 3) $(0, 2)$ 4) $(2, 4)$

33. The number of points on the curve $y = ||1 - e^x| - 2|$ from which two perpendicular tangents can be drawn to the parabola $x^2 = -4y$ is equal to

- 1) 2 2) 3 3) 4 4) 5

34. The equation to the Locus of the point of intersection of the perpendicular tangents to the parabola $x^2 - 4x + 4y - 8 = 0$ is

- 1) $y - 4 = 0$ 2) $x - 3 = 0$
 3) $x + y = 2$ 4) $x + 3 = 0$

35. The mirror image of the parabola $y^2 = 4x$ in the tangent to the parabola at the point $(1, 2)$ is

- 1) $(x-1)^2 = 4(y+1)$ 2) $(x+1)^2 = 4(y+1)$
 3) $(x+1)^2 = 4(y-1)$ 4) $(x-1)^2 = 4(y-1)$

36. A chord of parabola $y^2 = 4ax$ subtends a right angle at the vertex. The tangents at the extremities of chord intersect on

1. $x + a = 0$ 2. $x + 2a = 0$
 3. $x + 3a = 0$ 4. $x + 4a = 0$

37. The point on the parabola $y = x^2 + 7x + 2$ which is closest to the line $y = 3x - 3$ is

1. $(2, 8)$ 2. $(2, -8)$ 3. $(-2, 8)$ 4. $(-2, -8)$

38. If the distance of two points P and Q on the parabola $y^2 = 4ax$ from the focus of a parabola are 4 and 9 respectively then the distance of the point of intersection of tangents at P and Q from the focus is

1. 8 2. 6 3. 5 4. 13

39. Point of contact of the line $kx + y - 4 = 0$ w.r.t the parabola $y = x - x^2$ is

1. $(-2, 2)$ 2. $(2, -2)$ 3. $(-2, 6)$ 4. $(2, -6)$

NORMAL AND ITS APPLICATIONS

40. If a normal subtends a right angle at the vertex of a parabola $y^2 = 4ax$ then its length is

1. $2\sqrt{3}a$ 2. $\sqrt{3}a$ 3. $6\sqrt{3}a$ 4. $8\sqrt{3}a$

41. $y = x\sqrt{2} - 8\sqrt{2}$ is a normal chord to $y^2 = 8x$. Then its length is

- 1) $16\sqrt{3}$ 2) $4a\sqrt{3}$ 3) $12\sqrt{3}$ 4) $2a\sqrt{3}$

42. A normal chord of the parabola $y^2 = 4x$ makes an angle 45° with the axis of the parabola. Then its length is

- 1) $8\sqrt{2}$ 2) $10\sqrt{2}$ 3) $6\sqrt{3}$ 4) $4\sqrt{3}$

43. The point of intersection of normals to the parabola $y^2 = 4x$ at the points whose ordinates are 4 and 6 is

- 1) $(30, -21)$ 2) $(21, -30)$
 3) $(17, -19)$ 4) $(19, -18)$

44. The set of points on the axis of the parabola $y^2 - 2y - 4x + 5 = 0$ from which all the three normals to the parabola are real is :

- 1) $\{(x, 1); x \geq 3\}$ 2) $\{(x, -1); x \geq 1\}$
 3) $\{(x, 3); x \geq 1\}$ 4) $\{(x, -3); x \geq 3\}$

45. The normal at $P(8, 8)$ to the parabola $y^2 = 8x$ cuts it again at Q then PQ =

- 1) 10 2) $10\sqrt{5}$ 3) $5\sqrt{10}$ 4) 50

PROBLEMS ON CHORD

46. The length of the chord to the parabola $x^2 = 4ay$ passing through the vertex and having slope $\tan \alpha$ is

1. $4a \cot \alpha \sec \alpha$ 2. $4a \tan \alpha \sec \alpha$
 3. $4a \cos \alpha \cot \alpha$ 4. $4a \sin \alpha \tan \alpha$

47. An equilateral triangle is inscribed in the parabola $y^2 = 4ax$ where one vertex is at the vertex of the parabola. The length of side of triangle
- 1) $8\sqrt{3}a$ 2) $4\sqrt{3}a$
 3) $3\sqrt{3}a$ 4) $2\sqrt{3}a$
48. A line L passing through the focus of the parabola $y^2 = 4(x-1)$ intersects the parabola in two distinct points. If 'm' be the slope of the line L, then :
- 1) $m \in R$ 2) $-1 < m < 1$
 3) $m < -1$ or $m > 1$ 4) $m \neq 0$
49. Let PQ be a chord of the parabola $y^2 = 4x$. A circle drawn with PQ as diameter passes through the vertex 'V' of the parabola. If the area of triangle PVQ is 20 then coordinates of P are
- 1) $(-16, -8)$ 2) $(-16, 8)$ 3) $(16, -8)$ 4) $(8, 16)$
50. The triangle PQR of area 'A' is inscribed in the parabola $y^2 = 4ax$ such that the vertex P lies at the vertex of the parabola and base QR is a focal chord. The modulus of the difference of the ordinates of the points Q and R is
- 1) $\frac{A}{2a}$ 2) $\frac{A}{a}$ 3) $\frac{2A}{a}$ 4) $\frac{4A}{a}$
51. AB is a chord of the parabola $y^2 = 4ax$ with vertex A. BC is drawn $\perp$ as to AB meets the axis at C. The projection of BC on the axis of the parabola is
- 1) a 2) 2a 3) 4a 4) 8a
52. Consider the parabola $y^2 = 4x$; A $(4, -4)$; B $(9, 6)$ be two fixed points on the parabola. Let 'C' be a moving point on the parabola between A and B such that the area of the triangle ABC is maximum. The coordinate of C is
- 1) $\left(\frac{1}{4}, 1\right)$ 2) $(4, 4)$
 3) $(3, 2\sqrt{3})$ 4) $(3, -2\sqrt{3})$
53. Minimum area of circle which touches the parabola $y = x^2 + 1$ and $y^2 = x - 1$ is
- 1) $\frac{9\pi}{16}$ sq units 2) $\frac{9\pi}{32}$ sq units
 3) $\frac{9\pi}{8}$ sq units 4) $\frac{9\pi}{4}$ sq units
54. A ray of light travels along a line $y=4$ and strikes the surface of a curves $y^2 = 4(x+y)$, then equation of the line along which reflected ray travel is
- 1) $x=0$ 2) $x=2$ 3) $x+y=4$ 4) $2x+y=4$
55. PSQ is a focal chord of a parabola whose focus is S and Vertex A, PA and QA are produced to meet the directrix in R and T respectively. Then $\angle RST =$
- 1) 90° 2) 60° 3) 45° 4) 30°
56. The tangents to the parabola $y^2 = 4ax$ at $P(t_1)$ and $Q(t_2)$ intersect at R. Then the area of ΔPQR is
- 1) $\frac{a^2}{2}(t_1 - t_2)^2$ 2) $\frac{a^2}{2}(t_1 - t_2)$
 3) $\frac{a^2}{2}(t_1 - t_2)^3$ 4) $a^2(t_1 - t_2)^2$
57. The circle $x^2 + y^2 + 2\lambda x = 0$, $\lambda \in R$ touches the parabola $y^2 = 4x$ externally, then
- 1) $\lambda > 0$ 2) $\lambda < 0$ 3) $\lambda > 1$ 4) $\lambda = 0$
58. The cable of a uniformly loaded suspension bridge hangs in the form of a parabola. The road way which is horizontal and 100m long is supported by vertical wires attached to the cable, the longest wire being 30m and the shortest being 6m. The length of a supporting wire attached to the road way 18m from its middle
- 1) 10m 2) 9.11m 3) 9m 4) 11m
59. M is the foot of the perpendicular from a point P on the parabola $y^2 = 8(x-3)$ to its directrix and S is the focus of the parabola, if SPM is an equilateral triangle, the length of each side of the triangle is
- 1) 2 2) 3 3) 4 4) 8

60. $A=(-2, 0)$ and P is a point on the parabola $y^2 = 8x$. If Q bisects $\overline{AP}$ and the locus of Q is a parabola then its focus is

- 1) $(0, 0)$ 2) $(1, 1)$ 3) $(5, 0)$ 4) $(4, 0)$

61. If the angular bisectors of the coordinate axes cut the parabola $y^2 = 4ax$ at the points O, A, B then the area of $\triangle OAB$ is (O is the origin)

- 1) $32a^2$ 2) $16a^2$ 3) $64a^2$ 4) $8a^2$

62. For the parabola $y^2 = 8x$ tangent and normal are drawn at $P(2, 4)$ which meet the axis of the parabola in A and B . Then the length of the diameter of the circle through A, P, B is

- 1) 2 2) 4 3) 8 4) 6

63. A tangent to $y^2 = 4ax$ meets x axis at T and tangent at vertex A in P and the rectangle $TAPQ$ is completed. Then the locus of Q is given by

- 1) $y^2 + 4ax = 0$ 2) $y^2 + 2ax = 0$
3) $y^2 = 2ax$ 4) $y^2 + ax = 0$

64. If the tangent at P on $y^2 = 4ax$ meets the tangent at the vertex in Q , and S is the focus of the parabola, then $\angle SQP =$

- 1) $\frac{\pi}{3}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{2}$ 4) $\frac{2\pi}{3}$

65. Locus of the point of intersection of tangents at the end points of normal chord of the parabola $y^2 = 4ax$ is

- 1) $xy^2 + 2a(2a^2 + y^2) = 0$ 2) $xy^2 = 2a(2a^2 + y^2)$
3) $xy^2 + a(2a^2 + y^2) = 0$ 4) $xy^2 = a(2a^2 + y^2)$

LEVEL-II (C.W.) - KEY

- | | | | | | |
|-------|-------|-------|-------|-------|-------|
| 1) 1 | 2) 3 | 3) 3 | 4) 2 | 5) 4 | 6) 3 |
| 7) 3 | 8) 4 | 9) 2 | 10) 2 | 11) 3 | 12) 1 |
| 13) 2 | 14) 4 | 15) 3 | 16) 1 | 17) 2 | 18) 4 |
| 19) 2 | 20) 2 | 21) 1 | 22) 2 | 23) 4 | 24) 4 |
| 25) 1 | 26) 2 | 27) 2 | 28) 1 | 29) 1 | 30) 1 |
| 31) 3 | 32) 1 | 33) 1 | 34) 1 | 35) 3 | 36) 4 |
| 37) 4 | 38) 2 | 39) 2 | 40) 3 | 41) 3 | 42) 1 |
| 43) 2 | 44) 1 | 45) 2 | 46) 2 | 47) 1 | 48) 4 |
| 49) 3 | 50) 3 | 51) 3 | 52) 1 | 53) 2 | 54) 1 |
| 55) 1 | 56) 3 | 57) 1 | 58) 2 | 59) 4 | 60) 1 |
| 61) 2 | 62) 3 | 63) 4 | 64) 3 | 65) 1 | |

LEVEL-II (C.W.) - HINTS

1. Given points are end points of latusrectum. Focus is $(\alpha, \beta + a) = (-1, 1)$, $a = \pm 1$

vertices $(-1, 0)$ & $(-1, 2)$

$\therefore$ equation of parabolas are $(x+1)^2 = 4y$ and

$$(x+1)^2 = 4(y-2)$$

2. $SP^2 = PM^2$

d is the perpendicular distance from the focus to the directrix

$\therefore$ The length of the latusrectum = $2l$

$$2 \times \left| \frac{5(1) - 12(3) + 17}{\sqrt{25 + 144}} \right| = \frac{2}{13} \times 14 = \frac{28}{13}$$

3. $\tan 30^\circ = \frac{y}{x} \Rightarrow x = \sqrt{3}y$

substituting in the given parabola $y^2 = 8x$

$$\Rightarrow y = 8\sqrt{3}$$

then length of double ordinate = $16\sqrt{3}$

4. Given point $P(3, 0)$

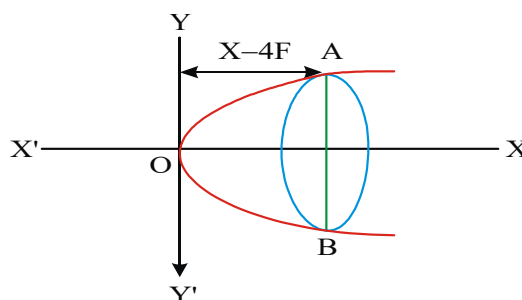
Equation of the chord AB is $S_1 = 0 \Rightarrow 3x - 1 = 0$

Let $T(x, y)$ point on the locus then $TP = TM$

$$\sqrt{(x-3)^2 + y^2} = \left| \frac{3x-1}{3} \right|$$

squaring and simplify

5. Here $a=5$; Origin is taken as vertex and axis of the mirror lies along the positive X -axis.



the equation of parabola is $y^2 = 4(5)x = 20x$.

when $x=45$; $y = \pm 30$; $AB=2y=60\text{cm}$.

6. Length of double ordinate is $4\sqrt{aK}$; $a = \frac{1}{4}$ (length of latusrectum); $K =$ given distance

7. Equation of the parabola is of the form $SP^2 = PM^2$

$$(x-5)^2 + (y-3)^2 = \left(\frac{3x-4y+1}{5}\right)^2$$

Focus is $(5, 3)$ directrix is $3x-4y+1=0$

Hence axis is $4x+3y=20+9=29$

8. Given $y = x^2 + 2px + 13$

$$\Rightarrow y - (13 - p^2) = (x + p)^2$$

Now, vertex is at $(-p, 13 - p^2) \equiv (-p, 4)$

$$\Rightarrow 13 - p^2 = 4 \Rightarrow p^2 = 9 \Rightarrow p = \pm 3$$

9. $(h, k) = (2, 0)$ and $s = (4, 0)$

10. Common chord of the given two circles

$$S = 0 \text{ \& } S^1 = 0 \text{ is } S - S^1 = 0 \Rightarrow 3x - y - 2 = 0$$

Vertex $(0, 0)$ is given

'Z' is foot of perpendicular of $A(0, 0)$ on the

directrix $\Rightarrow Z = \left(\frac{3}{5}, -\frac{1}{5}\right)$. Since A is the midpoint

$$\text{of } SZ \Rightarrow \text{focus } S = \left(-\frac{3}{5}, \frac{1}{5}\right)$$

Directrix $x=0 \Rightarrow -a + h = 0 \Rightarrow a = 2$

11. Take the equation of the parabola in the form (perpendicular distance from $p(x, y)$ to the axis)² = (length of latusrectum)(perpendicular distance from $p(x, y)$ to the tangent at the vertex)

12. The general equation of a parabola having its axis

parallel to y - axis is $y = ax^2 + bx + c$ (i)

This is touched by the line $y = x$ at $x = 1$

Therefore, slope of the tangent at $(1, 1)$ is 1 and

$x = ax^2 + bx + c$ must have equal roots

$$\Rightarrow \left(\frac{dy}{dx}\right)_{(1,1)} = 1 \text{ and } (b-1)^2 = 4ac$$

$$\Rightarrow 2a + b = 1 \dots\dots(2)$$

$$\text{and } (b-1)^2 = 4ac \dots\dots(3)$$

Also, $(1, 1)$ lies on equation (i)

$$\Rightarrow a + b + c = 1 \dots\dots\dots(4)$$

From (2) and (4) $a - c = 0 \Rightarrow a = c$

Then from $a + b + c = 1, 2c + b = 1$

$$\Rightarrow 2f(0) + f'(0) = 1$$

$$[\because f(0) = c \text{ and } f'(0) = b]$$

13. $S(1, 2), Z(2, 3)$; Directrix $y - 3 = -1(x - 2)$,

Now use $SP^2 = PM^2$

14. If $P \equiv (x, y)$ then

$$\sqrt{x^2 + y^2} < |a - x|, \sqrt{x^2 + y^2} < |a + x|$$

$$\sqrt{x^2 + y^2} < |a - y|, \sqrt{x^2 + y^2} < |a + y|$$

Squaring them we get the region bounded by the

curves $x^2 + y^2 = (a - x)^2, x^2 + y^2 = (a + x)^2$

$$x^2 + y^2 = (a - y)^2, x^2 + y^2 = (a + y)^2$$

15. On observation the given points are collinear. Hence number of parabolas = 0

16. Eliminating 't' and x is always positive

17. Equation of the tangent line $y = mx + \frac{9}{m}$

It meets the axes in the points $\left(-\frac{9}{m^2}, 0\right) = A$

$$\left(0, \frac{9}{m}\right) = B \text{ and } O = (0, 0)$$

The centroid of triangle OAB =

$$\left(-\frac{3}{m^2}, \frac{3}{m}\right) = (x, y) \Rightarrow \text{eliminating } m, \text{ we get the}$$

locus

18. Verification

19. solving $x + y = K + 1$ and

$$y = x(1 - x) \Rightarrow x^2 - 2x + K + 1 = 0 \text{ and } \Delta = 0$$

20. $m_1 + m_2 = \frac{y_1}{x_1}, m_1 \cdot m_2 = \frac{a}{x_1}$ But $m_1 = 2m_2$

$$3m_2 = \frac{y_1}{x_1}, 2m_2^2 = \frac{a}{x_1} \text{ eliminate } m_2$$

21. Use $a^{1/3}x + b^{1/3}y + (ab)^{2/3} = 0$

22. $\tan \theta = \frac{\sqrt{S_{11}}}{x_1 + a}$;

$$\theta = 45^\circ = x_1^2 + 2ax_1 + a^2 = y_1^2 - 4ax_1$$

locus is $x^2 - y^2 + 6ax + a^2 = 0$

23. $\Delta = \frac{(S_{11})^{3/2}}{2a}$

24. $x + a + b = 0$

25. Select the equation satisfying the point $[2at^2, 3at]$

26. $m_1^2 + m_2^2 = k \Rightarrow (m_1 + m_2)^2 - 2m_1m_2 = k$

27. Verification

28. $y = mx + \frac{3}{m}$ passing through $(1, 4)$

29. Origin lie in directrix

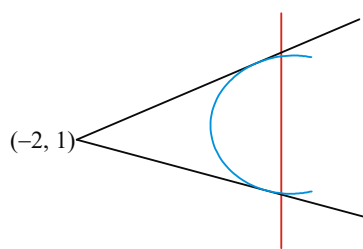
30. $\sec^2 \theta_1 \sec^2 \theta_2 = \frac{1}{k^2}$; $(1 + \tan^2 \theta_1)(1 + \tan^2 \theta_2) = \frac{1}{k^2}$

$$[1 + (\tan \theta_1 + \tan \theta_2)^2 - 2 \tan \theta_1 \tan \theta_2 + \tan^2 \theta_1 \tan^2 \theta_2] = \frac{1}{k^2}$$

$$\left[1 + \frac{y_1^2}{x_1^2} - \frac{2a}{x_1} + \frac{a^2}{x^2}\right] = \frac{1}{k^2}; \quad k^2[(x_1 - a)^2 + y_1^2] = x_1^2$$

equation to locus is $x^2 = k^2[(x - a)^2 + y^2]$

31.



Select the point satisfying the equation of chord of contact of $(-2, 1)$ w.r.to $y^2 = 4x$

32. Let P is the required point, then P lies on directrix $x = -2$, hence $P = (-2, 0)$

33. Mutually perpendicular tangents can be drawn from a point on the directrix. Solving the directrix $y = 1$ with the curve $y = |1 - e^x| - 2$ we get two point of intersections.

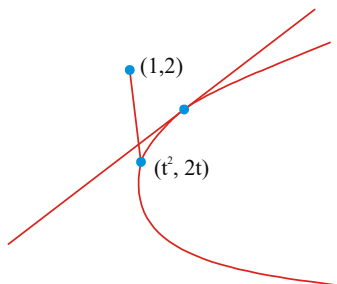
34. Given equation $(x - 2)^2 = -4(y - 3)$ vertex

$$(\alpha, \beta) = (2, 3); \quad 4a = -4, a = -1$$

$\therefore$ equation of directrix $y = \beta - a$

$$\Rightarrow y - 4 = 0 \Rightarrow y = 3 + 1 \Rightarrow y = 4$$

35.



Any point on the given parabola is $(t^2, 2t)$

The equation of the tangent at $(1, 2)$ is $x - y + 1 = 0$

The image (h, k) of the point $(t^2, 2t)$ in $x - y + 1 = 0$ is

$$\text{given by } \frac{h - t^2}{1} = \frac{k - 2t}{-1} = -2 \left(\frac{t^2 - 2t + 1}{1 + 1} \right)$$

$$\Rightarrow h = t^2 - t^2 + 2t - 1$$

$$k = t^2 + 1 \therefore k = t^2 + 1 \therefore \text{Elimination } t \text{ we get}$$

$$(h + 1)^2 = 4(k - 1) \text{ reflection is } (x + 1)^2 = 4(y - 1)$$

36. Apply $t_1 t_2 = -4$ and point of intersection of the tangents at t_1, t_2 is $(at_1 t_2, a(t_1 + t_2)) \Rightarrow x = -4a$ satisfies.

37. Differentiating the given curve w.r.t to x

$$2x + 7 = 3 = \text{slope of the tangent, here } x = -2$$

put in the given curve, we get $y = -8$

$\therefore$ the point is $(-2, -8)$.

38. If T is the point of the intersection of the tangents at P, Q. We have $ST^2 = SP \cdot SQ$

39. $4 - kx = x - x^2 \Rightarrow x^2 - (k + 1)x + 4 = 0$

Equation has coincident roots $\Rightarrow$ discriminant is zero. $\therefore k = 3$ or -5 .

The points are $(2, -2)$ or $(-2, 6)$, but $(-2, 6)$ does not satisfy the given curve.

40. Apply the condition $t = \pm\sqrt{2}$

$$\text{length} = 4a \frac{(1+t^2)^{3/2}}{t^2}$$

41. $t = -\sqrt{2}$, $\text{length} = 4a \frac{(1+t^2)^{3/2}}{t^2}$

42. Slope $t = -1$, $\text{length} = 4a \frac{(1+t^2)^{3/2}}{t^2}$

43. Use $\left[2a + a\left[(t_1 + t_2)^2 - t_1 t_2\right], -at_1 t_2(t_1 + t_2)\right]$

44. $(y-1)^2 = 4(x-1)$, Axis of parabola is $y=1$
equation of normal $(y-1) = m(x-1) - 2m - m^3$
let $(h, 1)$ is point on its axis then
 $0 = m(h-1) - 2m - m^3 \Rightarrow m^3 = h-3 \Rightarrow h \geq 3$
for real values of m

45. Apply $t_2 = -t_1 - \frac{2}{t_1}$

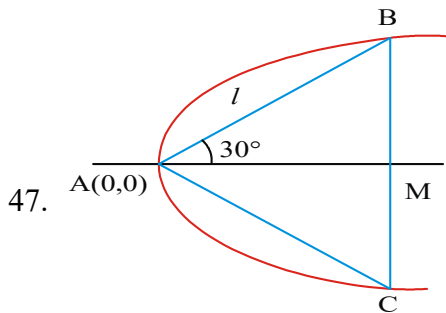
46. Let AP be a chord of the parabola $x^2 = 4ay$.

slope of AP = $\tan \alpha$, $P = (2at, at^2)$

slope of AP = $\frac{at^2}{2at} = \frac{t}{2} = \tan \alpha$

$$\Rightarrow AP = \sqrt{(2at)^2 + (at^2)^2} = at\sqrt{4+t^2}$$

put $t = 2 \tan \alpha$ then $AP = 4a \tan \alpha \sec \alpha$



47.

$$\cos 30^\circ = \frac{AM}{l}; \sin 30^\circ = \frac{BM}{l}$$

$$AM = l \left(\frac{\sqrt{3}}{2} \right); BM = l \left(\frac{1}{2} \right)$$

$$B = \left(\frac{l\sqrt{3}}{2}, \frac{l}{2} \right) \text{ lies on } y^2 = 4ax \Rightarrow l = 8\sqrt{3}a$$

48. The given parabola is $y^2 = 4(x-1)$

Shifting the origin to $(1,0)$ by putting $x-1=X$ and

$y=Y$ we get: $Y^2 = 4X$

Its focus is $(1, 0)$ i.e. $(2, 0)$

Any line through $(2,0)$ is $y=m(x-2)$

Putting the value of y from (2) in (1), we get

$$m^2(x-2)^2 = 4(x-1)$$

$$\Rightarrow m^2 x^2 - 4(m^2 + 1)x + 4(m^2 + 1) = 0$$

If $m \neq 0$, then $D = 16(m^2 + 1)^2 - 16(m^2 + 1)$

$$= 16(m^2 + 1) > 0 \text{ for all } m$$

But if $m = 0$, then x does not have real and distinct values. Hence $m \in R$ except $m = 0$

49. slope of $\overline{PV}$ X slope of $\overline{QV} = -1$;

$$P(at_1^2, 2at_1), Q(at_2^2, 2at_2) \Rightarrow t_1 t_2 = -4$$

$$\text{Area} : \frac{1}{2} |x_1 y_2 - x_2 y_1| = 20 \Rightarrow |t_1 t_2 (t_2 - t_1)| = 20$$

$$\Rightarrow \left| -4 \left(\frac{-4}{t_1} - t_1 \right) \right| = 20$$

$$\Rightarrow \frac{4}{t_1} + t_1 = 5; \frac{4}{t_1} + t_1 = -5$$

$\Rightarrow t_1^2 - 5t_1 + 4 = 0 \Rightarrow t_1 = 1; t_1 = 4$ then there is no answer in the given option.

$$\Rightarrow t_1^2 + 5t_1 + 4 = 0 \Rightarrow t_1 = -1; t_1 = -4$$

If we take $t_1 = -4$

$$\therefore P = (16, -8).$$

50. $P(0,0); Q(at_1^2, 2at_1); R(at_2^2, 2at_2)$

Since QR is a focal chord $t_1 t_2 = -1$

$$A = \frac{1}{2} |(at_1^2)(2at_2) - (at_2^2)(2at_1)|$$

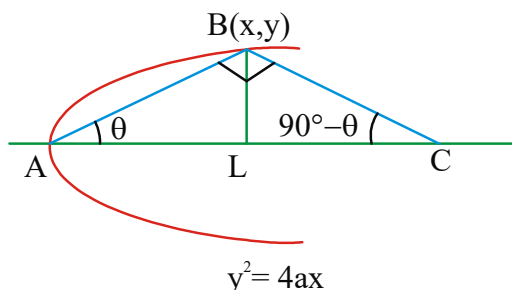
$$= |a^2 t_1 t_2 (t_1 - t_2)| = |a^2 (-1)(t_1 - t_2)| = A = a^2 |t_2 - t_1|$$

The difference of the ordinates of point Q and R is

$$|2a(t_2 - t_1)|; \frac{2A}{a} = \left| 2a \frac{A}{a^2} \right|$$

51. From $\triangle BLC$ $\tan(90^\circ - \theta) = \frac{BL}{LC}$

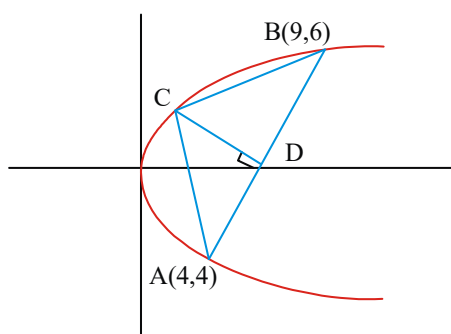
$$\Rightarrow LC = \frac{BL}{\cot \theta} = BL \tan \theta = y \tan \theta$$



$\triangle BLC$ $\tan(90^\circ - \theta) = \frac{BL}{LC}$

$$\Rightarrow LC = \frac{BL}{\cot \theta} = BL \tan \theta = y \tan \theta$$

But $\tan \theta = \frac{y}{x} \Rightarrow \frac{y \cdot y}{x} = \frac{4ax}{x} = 4a$



52.

Area of triangle ABC is maximum if CD is maximum
tangent drawn to parabola at C should be parallel

to AB. slope of $AB = \frac{6+4}{9-4} = 2$.

$$y^2 = 4x \Rightarrow \frac{dy}{dx} = \frac{2}{y} = 2 \Rightarrow y = 1$$

$$\therefore x = \frac{1}{4} \text{ Point is } \left(\frac{1}{4}, 1\right)$$

53. Given parabolas are symmetric about $y=x$. tangent at point A is parallel to $y=x$.

$$\Rightarrow \frac{dy}{dx} = 2x \Rightarrow 2x = 1 \Rightarrow x = \frac{1}{2}; y = \frac{5}{4}$$

$$A\left(\frac{1}{2}, \frac{5}{4}\right) B\left(\frac{5}{4}, \frac{1}{2}\right)$$

$$\text{Radius} = \frac{1}{2} AB = \frac{1}{2} \sqrt{\frac{9}{16} + \frac{9}{16}} = \frac{3}{8} \sqrt{2}$$

$$\text{Area} = \frac{9\pi}{32}$$

54. Given curve is $(y-2)^2 = 4(x+1)$

vertex = $(\alpha, \beta) = (-1, 2)$; Focus = $S = (0, 2)$

P is point on the curve and $y=4$ is $(0, 4)$

reflected ray passes through focus $S=(0, 2)$

equation of the line along which reflected ray travels is $x = 0$

55. Let $P \equiv (at_1^2, 2at_1), Q \equiv (at_2^2, 2at_2) \Rightarrow t_1 t_2 = -1$

Equation of AP is, $y = \frac{2}{t_1} x \Rightarrow R \equiv (-a, -2a/t_1)$

Similarly, $T \equiv \left(-a, \frac{-2a}{t_2}\right)$

$$\text{Slope of RS} = \frac{-2a/t_1}{2a} = -\frac{1}{t_1}$$

$$\text{Slope of TS} = \frac{-2a/t_2}{2a} = -\frac{1}{t_2}$$

$$\text{Now, } \left(-\frac{1}{t_1}\right) \times \left(-\frac{1}{t_2}\right) = \frac{1}{t_1 t_2} = -1 \Rightarrow \angle RST = \frac{\pi}{2}$$

56. Remember as a formula

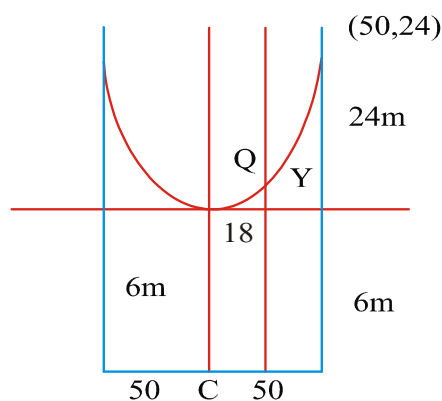
57. Given, equations are $x^2 + y^2 + 2\lambda x = 0$

$y^2 = 4x$ centre of circle is $(-\lambda, 0)$ and radius is

$$\sqrt{\lambda^2 + 0} = \lambda$$

Clearly the circle will touch the parabola externally if centre of circle lies on $-ve$ x -axis.

58. Parabola is $x^2 = 4ay$.



$$2500 = 4a(24) \Rightarrow 4a = \frac{625}{6}$$

$$\therefore x^2 = \frac{625}{6} y \text{ point.}$$

Q(18,y) lies on

$$x^2 = 4ay \Rightarrow 18^2 = \frac{625}{6} y \Rightarrow y = 3.11m \text{ length of}$$

supports wire is attached = $6 + 3.11 \Rightarrow 9.11m$

59. Since SPM is an equilateral triangle $\angle SMP = 60^\circ$
 $\Rightarrow \angle SMZ = 30^\circ$. From triangle SZM

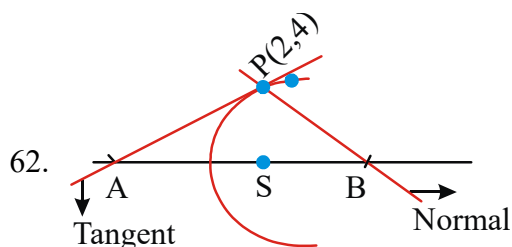
$$\frac{SZ}{SM} = \sin 30^\circ \Rightarrow SM = 2SZ \quad SM = 2 \times 4 = 8$$

Hence length of each side of the triangle is 8

60. $P = (at^2, 2at)$, $A = (-2, 0)$, midpoint of

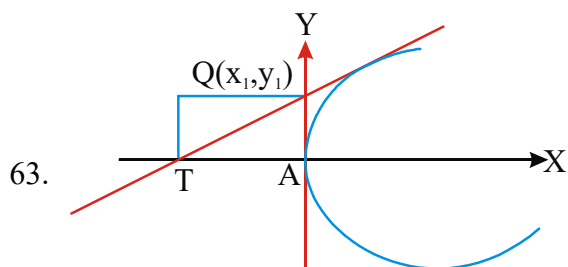
$AP = Q(x, y)$. Hence eliminate 't'

61. Angular bisectors $y = x$, $y = -x$
 Solve and find A, B



$$SA = SP = SB; \quad SP = 2 + 2 = 4$$

AB is the diameter of the circle through A, P, B and S is centre. $\therefore \overline{AB} = 8$



let $Q = (x_1, y_1)$ equation of the line TP is $\frac{x}{x_1} + \frac{y}{y_1} = 1$

$$xy_1 + yx_1 - x_1y_1 = 0$$

This is a tangent to $y^2 = 4ax$

$$\text{condition of tangency } am^2 = \ell n \Rightarrow ax_1^2 = y_1 \times -x_1y_1$$

$$\text{equation to locus } y^2 + ax = 0$$

64. Let $P(at^2, 2at)$ be any point on the parabola

$$y^2 = 4ax$$

Equation of tangent at P is $2aty = 2a(x + at^2)$

$$\Rightarrow y = \frac{1}{t}x + at$$

Intersection of this tangent with the tangent to parabola at vertex i.e. y-axis is Q(0, at)

$$\text{Now } \angle SPQ = \tan^{-1} \left| \frac{\frac{1}{t} - at}{1 + \frac{1}{t} \left(\frac{at}{-a} \right)} \right|$$

$$\Rightarrow \angle SPQ = \tan^{-1}(\infty) = \frac{\pi}{2}$$

$$65. \quad x = at_1t_2 = at_1 \left(-t_1 - \frac{2}{t_1} \right) = -a(t_1^2 + 2)$$

$$y = a(t_1 + t_2) = a \left(t_1 - t_1 - \frac{2}{t_1} \right) = -\frac{2a}{t_1} \Rightarrow t_1 = -\frac{2a}{y}$$

$$\therefore \text{Required locus is } x = -a \left(\frac{4a^2}{y^2} + 2 \right)$$

$$\Rightarrow xy^2 + 2a(2a^2 + y^2) = 0$$



LEVEL-II (H.W)

FOCUS, VERTEX, DIRECTRIX, LATUSRECTUM, AXIS, FOCAL CHORD, FOCAL DISTANCE, DOUBLE ORDINATE

1. The parabola $y^2 = px$ passes through the point of intersection of the lines $\frac{x}{3} + \frac{y}{2} = 1$ and

$$\frac{x}{2} + \frac{y}{3} = 1 \text{ its focus is}$$

$$1) \left(\frac{3}{10}, 0 \right) \quad 2) \left(\frac{3}{5}, 0 \right) \quad 3) \left(\frac{3}{7}, 0 \right) \quad 4) \left(\frac{6}{7}, 0 \right)$$

2. If the equation $(2x - y - 3)^2 = -20(x + 2y - 4)$ represents a parabola then its vertex is

$$1) (3, 3) \quad 2) (6, -1) \quad 3) (2, 1) \quad 4) (4, 4)$$

3. The parabola $y^2 = 2ax$ passes through the centre of the circle $4x^2 + 4y^2 - 8x + 12y - 7 = 0$. Its directrix is

- 1) $4x + 9 = 0$ 2) $4x + 15 = 0$
3) $16x + 9 = 0$ 4) $16x - 7 = 0$

4. If the vertex of the parabola $y = x^2 - 8x + c$ lies on x axis then the value of c is

- 1) 12 2) 14 3) 8 4) 16

EQUATION OF PARABOLA/CONIC

5. A parabola has x-axis as its axis, y-axis as its directrix and $4a$ as its latusrectum. If the focus lies to the left side of the directrix then the equation of the parabola is

- 1) $y^2 = 4a(x + a)$ 2) $y^2 = 4a(x - a)$
3) $y^2 = -4a(x + a)$ 4) $y^2 = 4a(x + 2a)$

6. Two parabolas with the same axis, focus of each being exterior to the other and the latusrectum being $4a$ and $4b$. The locus of the middle points of the intercepts between the parabolas made on the lines parallel to the common axis is a

- 1) straight line if $a > b$ 2) parabola if $a \neq b$
3) parabola for all a, b 4) ellipse, if $b > a$

7. The equation

$$16x^2 + y^2 + 8xy - 74x - 78y + 212 = 0$$

represents

- 1) a circle 2) a parabola
3) an ellipse 4) a hyperbola

8. The equation of parabola whose latus rectum is 2 units, axis of line is $x + y - 2 = 0$ and tangent at the vertex is $x - y + 4 = 0$ is given by

- 1) $(x + y - 2)^2 = 4\sqrt{2}(x - y + 4)^2$
2) $(x - y - 4)^2 = 4\sqrt{2}(x + y - 2)$
3) $(x + y - 2)^2 = 2\sqrt{2}(x - y + 4)$
4) $(x - y - 4)^2 = 2\sqrt{2}(x - y + 2)^2$

9. If the vertex and focus of a parabola are $(2, 1)$ and $(1, -1)$ then its equation is

- 1) $4x^2 + y^2 - 4xy + 8x + 46y - 65 = 0$
2) $4x^2 + y^2 - 4xy - 8x - 46y + 72 = 0$
3) $4x^2 + y^2 - 4xy + 8x + 46y - 71 = 0$
4) $4x^2 + y^2 - 4xy - 10x - 23y + 75 = 0$

TANGENT AND ITS APPLICATIONS

10. If the line $y = x + 2a$ touches the parabola $y^2 = 4a(x + a)$ then the point of contact is

- 1) $(2a, 0)$ 2) (a, a) 3) $(0, 2a)$ 4) $(-a, a)$

11. Equation of the common tangent to the parabola $y^2 = 24x$ and the circle $x^2 + y^2 = 18$ is

- 1) $x + y + 6 = 0$ 2) $x + y + 4 = 0$
3) $2x + y - 9 = 0$ 4) $x + 2y - 8 = 0$

12. Let P be a variable point. From P, PQ and PR are tangents drawn to the parabola $y^2 = 4x$. If $\angle QPR$ is always 45° , then locus of P is

- 1) $x^2 - y^2 + 6x + 1 = 0$ 2) $x^2 + y^2 + 6x + 1 = 0$
3) $x^2 - y^2 - 6x + 1 = 0$ 4) $x^2 + y^2 - 6x + 1 = 0$

13. If the line $lx + my + n = 0$ touches the parabola $y^2 = 4a(x - b)$ then

- 1) $am^2 = l(n + b)$ 2) $am^2 = lnb^2$
3) $am^2 = bl + nl$ 4) $am^2 = bl^2 + nl$

14. P is a point, Two tangents are drawn from it to the parabola $y^2 = 4x$ such that the slope of one tangent is three times the slope of the other. The locus of P is

- 1) a straight line 2) a circle
3) a parabola 4) an ellipse

15. The equations of the tangents at the ends of latusrectum of $y^2 = 4ax$ are

- 1) $x \pm y - a = 0$ 2) $x \pm y + a = 0$
3) $x \pm y - 3a = 0$ 4) $x \pm y + 2a = 0$

16. The number of real tangents that can be drawn to the curve

$$y^2 + 2xy + x^2 + 2x + 3y + 1 = 0 \text{ from the point } (1, -2) \text{ is}$$

- 1) 1 2) 2 3) 0 4) 3

NORMAL AND ITS APPLICATIONS

17. The normal at $(a, 2a)$ on $y^2 = 4ax$ meets the curve again at $(at^2, 2at)$. Then the value of t is equal to

- 1) 1 2) 3 3) -1 4) -3

18. The normal at $(ap^2, 2ap)$ on $y^2 = 4ax$, meets the curve again at $(aq^2, 2aq)$ then

- 1) $p^2 + pq + 2 = 0$ 2) $p^2 - pq + 2 = 0$
3) $p^2 + pq - 2 = 0$ 4) $p^2 + pq + 1 = 0$

19. The tangent at $P(at^2, 2at)$ to the parabola $y^2 = 4ax$ intersects X axis at A and the normal at P meets it at B then area of triangle PAB is

- 1) $4a^2 |t| \sqrt{1+t^2}$ 2) $2a^2 |t| (1+t^2)$
3) $4a^2 |t| (1+t^2)$ 4) $\frac{2a^2 (1+t^2)}{|t|}$

CHORD, CHORD OF CONTACT, CHORD LENGTH, MID POINT OF A CHORD

20. If the segment intercepted by the parabola $y^2 = 4ax$ with the line $lx + my + n = 0$ subtends a right angle at the vertex then

- 1) $4al + n = 0$ 2) $4al + 4am + n = 0$
3) $4am + n = 0$ 4) $al + n = 0$

21. The locus of the mid point of the line segment joining the focus to a moving point on the parabola $y^2 = 4ax$ is another parabola with directrix

- 1) $x = -a$ 2) $x = -a/2$ 3) $x = 0$ 4) $x = a/2$

22. A chord PP' of a parabola cuts the axis of the parabola at O. The feet of the perpendiculars from P and P' on the axis are M and M' respectively. If V is the vertex then, VM, VO, VM' are in

- 1) A.P 2) G.P 3) H.P 4) A.G.P

23. An equilateral triangle is inscribed in the parabola $y^2 = 4ax$ with one vertex at the origin. The radius of circumcircle of the triangle is

- 1) $\frac{a}{2}$ 2) $\frac{3a}{2}$ 3) $4a$ 4) $8a$

24. If the line $y = \sqrt{3}x - 3$ cuts the parabola $y^2 = x + 2$ at P and Q and if A be the point $(\sqrt{3}, 0)$, then $|AP \cdot AQ|$ is

- 1) $\frac{2}{3}(\sqrt{3} + 2)$ 2) $\frac{4}{3}(\sqrt{3} + 2)$
3) $\frac{4}{3}(2 - \sqrt{3})$ 4) $2\sqrt{3}$

25. Let P be the point (1,0) and Q be a point on the locus $y^2 = 8x$. The locus of midpoint of PQ

- 1) $x^2 + 4y + 2 = 0$ 2) $x^2 - 4y + 2 = 0$
3) $y^2 - 4x + 2 = 0$ 4) $y^2 + 4x + 2 = 0$

26. The chord AB of the parabola $y^2 = 4ax$ cuts the axis of the parabola at C. If $A = (at_1^2, 2at_1)$, $B = (at_2^2, 2at_2)$ and

$AC : AB = 1 : 3$, then

- 1) $t_2 - 2t_1 = 0$ 2) $t_2 + 2t_1 = 0$
3) $t_2 - t_1 = 0$ 4) $t_2 + t_1 = 0$

27. If $L_1 L_2$ is the latus rectum of $y^2 = 12x$, P is any point on the directrix. Then the area of $\Delta PL_1 L_2 =$

- 1) 32 2) 18 3) 36 4) 16

28. The triangle formed by the latusrectum of a parabola and the tangents drawn at the ends of latusrectum is always

- 1) equilateral 2) isosceles
3) right angled 4) right angled isosceles

29. The number of points of intersection of $x^2 + y^2 = 4x$ with $y^2 = 4(x - 3)$

- 1) 1 2) 2 3) 3 4) 4

30. The length of the portion of the normal at (1,1) to $x^2 = y$ intercepted between the axes is

- 1) $\frac{5\sqrt{3}}{2}$ 2) $\frac{3\sqrt{5}}{2}$ 3) $\frac{5\sqrt{3}}{4}$ 4) $\frac{3\sqrt{5}}{4}$

31. The tangents to a parabola at the vertex V and any point P meet at Q. If S be the focus then SP, SQ, SV are in

- 1) A.P 2) G.P 3) H.P 4) A.G.P

32. The point $(2a, a)$ lies inside the region bounded by the parabola $x^2 = 4y$ and its latus rectum. Then,

- 1) $0 \leq a \leq 1$ 2) $0 < a < 1$
3) $a > 1$ 4) $a < 0$

33. Perpendiculars are drawn on a tangent to the parabola $y^2 = 4ax$ from the points $(a \pm k, 0)$. The difference of their squares is

- 1) 4 2) 4a
3) 4k 4) 4ak

Let $P(\alpha, \beta)$ be the mid-point of AB. Then,

$$2\alpha = \frac{h^2}{4} \left(\frac{1}{a} - \frac{1}{b} \right) + l - m \text{ and } \beta = h$$

$$\Rightarrow 2\alpha = \frac{\beta^2}{4} \left(\frac{1}{a} - \frac{1}{b} \right) + l - m$$

Thus, the locus of (α, β) is

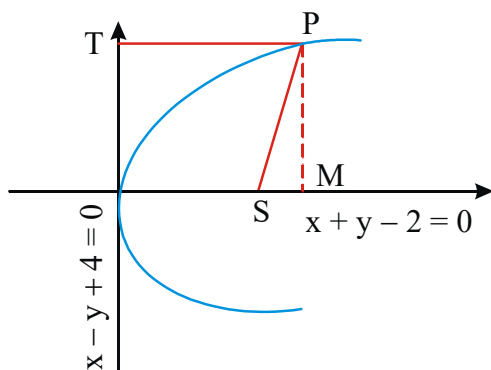
$$2x = \frac{y^2}{4} \left(\frac{1}{a} - \frac{1}{b} \right) + l - m$$

Clearly, it represents a parabola if $a \neq b$.

If $a = b$, then we have $2x = l - m$ which represents a line.

7. $\Delta \neq 0$ & $h^2 = ab$, parabola

8. Let $P(x, y)$ be any point on the parabola and say PM & PT are perpendicular from P on the axis & tangent at the vertex



or $(PM)^2 = \text{latus rectum (PT)}$

$$\therefore \left(\frac{x + y - 2}{\sqrt{1^2 + 1^2}} \right)^2 = 2 \frac{(x - y + 4)}{\sqrt{2}}$$

$$\Rightarrow (x + y - 2)^2 = 2\sqrt{2}(x - y + 4)$$

9. Find the directrix and use $SP^2 = PM^2$

$$12. \tan \theta = \frac{\sqrt{S_{11}}}{|x_1 + a|}$$

13. change the origin to point $(b, 0)$

$$x = X + b \Rightarrow X = x - b; y = Y + 0 \Rightarrow Y = y$$

$$\therefore l(X + b) + m(Y) + n = 0$$

The touch parabola $Y^2 = 4aX$

$$\Rightarrow \text{apply condition} \Rightarrow am^2 = l(bl + n)$$

14. $m_1 = m, m_2 = 3m$ and eliminate m .

Thus, the locus is $3y^2 = 16x$, which is a parabola

15. $L(a, 2a), L^1(a, -2a)$ tangents $S_1 = 0$

16. Equation of curve is $(x + y)^2 = -(2x + 3y + 1)$

Since second degree terms form a perfect square, therefore the curve is a parabola. Also

$$(-2)^2 + 2(1)(-2) + (1)^2 + 2(1) + 3(-2) + 1 = -2 < 0$$

$\therefore P(1, -2)$ lies inside the parabola

Hence, no real tangent can be drawn to the parabola

17. equation of the normal $y + xt = 2at + at^3$

$(a, 2a) = (at_1^2, 2at_1) \Rightarrow t_1 = 1$ it is passes through $(at^2, 2at)$

18. Since the normal at $(aq^2, 2ap)$ on $y^2 = 4ax$ meets the curve again at $(aq^2, 2aq)$, therefore

$$px + y = 2ap + ap^3 \text{ passes through } (aq^2, 2aq)$$

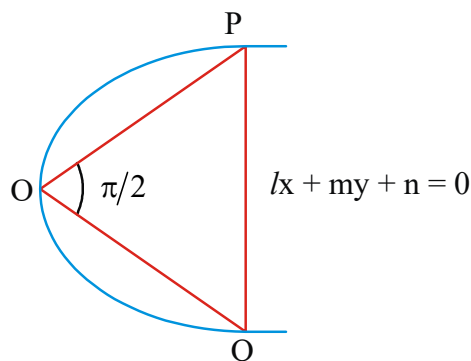
$$\Rightarrow paq^2 + 2aq = 2ap + ap^3$$

$$\Rightarrow p(q^2 - p^2) = 2(p - q) \Rightarrow p^2 + pq + 2 = 0$$

19. PA=length of tangent $= |2at|\sqrt{1+t^2}$

$$pb = \text{length of normal} = |2a|\sqrt{1+t^2}$$

$$\text{Area of PAB} = \frac{1}{2}(PA \cdot PB) = 2a^2 |t|(1+t^2)$$



20.

Combined equation of $\overline{OP}$ and $\overline{OQ}$ is

$$y^2 + 4ax \left(\frac{lx + my}{n} \right) = 0; \because \angle POQ = \frac{\pi}{2}; 4al + n = 0$$

21. If (h, k) is the mid point of line joining focus $(a, 0)$ and $Q(at^2, 2at)$ on parabola then

$$h = \frac{a + at^2}{2}, k = at$$

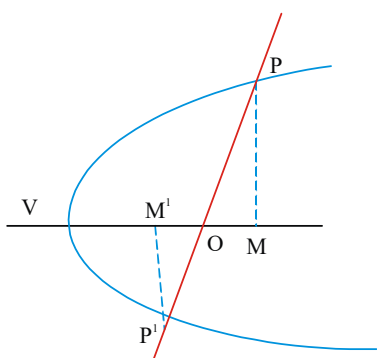
Eliminating 't' we get, $2h = a + a\left(\frac{k^2}{a^2}\right)$

$$\Rightarrow k^2 = a(2h - a) \Rightarrow k^2 = 2a(h - a/2)$$

Locus of (h, k) is $y^2 = 2a(x - a/2)$

whose directrix is $x - \frac{a}{2} = \frac{-a}{2} \Rightarrow x = 0$

22. Let the parabola be $y^2 = 4ax$ and $P(at_1^2, 2at_1)$



$\therefore VM = at_1^2$; $VM' = at_2^2$ and $VO = k$, we have

$$\begin{vmatrix} at_1^2 & 2at_1 & 1 \\ at_2^2 & 2at_2 & 1 \\ k & 0 & 1 \end{vmatrix} = 0 \quad (\text{since } P, P', O \text{ are collinear})$$

$$\Rightarrow k = at_1t_2 = 0$$

$$\therefore VM \cdot VM' = (at_1t_2)^2 = k^2 = OV^2$$

23 $R = \frac{\text{length of side}}{\sqrt{3}} = \frac{8\sqrt{3}a}{\sqrt{3}} = 8a$

24. The point $A(\sqrt{3}, 0)$ lies on the given line whose slope is $\sqrt{3}$. $\therefore \tan \theta = \sqrt{3} = \tan 60^\circ$; $\therefore \theta = 60^\circ$
Equation of line through A is

$$\frac{x - \sqrt{3}}{\cos 60^\circ} = \frac{y - 0}{\sin 60^\circ} = r$$

where r is the distance of any point from A in the direction of 60° . It intersects the parabola

$$y^2 = x + 2 \Rightarrow \left(\frac{r\sqrt{3}}{2}\right)^2 = \frac{r}{2} + \sqrt{3} + 2$$

$$\Rightarrow 3r^2 - 2r - 4(2 + \sqrt{3}) = 0$$

$$\Rightarrow r_1r_2 = -\frac{4(2 + \sqrt{3})}{3} = AP \cdot AQ$$

$$\therefore AP \cdot AQ = -\frac{4}{3}(\sqrt{3} + 2)$$

25. $P = (1, 0)$ $Q(h, k)$ such that $k^2 = 8h$

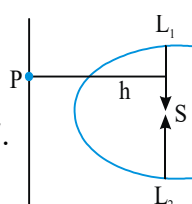
Let (α, β) be mid point of PQ

$$\therefore \alpha = \frac{h+1}{2}, \beta = \frac{k+0}{2}, 2\alpha - 1 = h, 2\beta = k$$

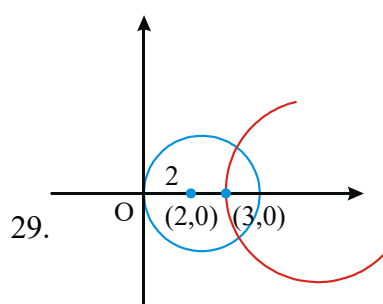
$$\therefore (2\beta)^2 = 8(2\alpha - 1) \Rightarrow \beta^2 = 4\alpha - 2$$

$$\Rightarrow y^2 - 4x + 2 = 0$$

26. $(C \div \overline{AB}) = 1 : 2$

27.  Area = $\frac{1}{2} L_1 L_2 \cdot h = \frac{1}{2} \times 12 \times 6 = 36$

28. Tangents meet on directrix perpendicularly. it is right angled Isosceles triangle



29.

Points of intersection=2

30. Normal at $(1, 1)$ is $x + 2y - 3 = 0 \rightarrow (1)$

Parabola $x^2 = y \rightarrow (2)$

Solving (1) & (2) $A(3, 0), B(0, 3/2)$; $\therefore AB = \frac{3\sqrt{5}}{2}$

31. Let the parabola be $y^2 = 4ax$. Q is the intersection of the lines $x=0$ and $ty = x + at^2$, where $P(at^2, 2at)$.

Solving these $Q=(0, at)$. Also, $S=(a, 0)$

$$\therefore SP^2 = \{a(t^2 - 1)\}^2 + 4a^2t^2 = a^2(t^2 + 1)^2$$

$$\therefore SQ^2 = a^2 + a^2t^2 = a^2(t^2 + 1) \text{ and } SV = a$$

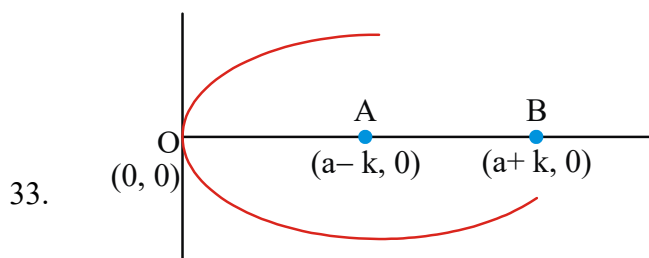
$$\therefore SQ^2 = SP \cdot SV$$

32. Let $S \equiv x^2 - 4y = 0$ since the point $(2a, a)$ lies inside the parabola,

$$\therefore S(2a, a) = 4a^2 - 4a < 0 \text{ or } a(a-1) < 0$$

Also, the vertex $A(0,0)$ and the point $(2a, a)$ are on the same side of the line $y=1$ (the equation of latus rectum) So, $a-1 < 0$ i.e. $a < 1$

From (1) and (2) we have $a(a-1) < 0$ or $0 < a < 1$



Tangent at the vertex

$$AO = a - k \quad BO = a + k$$

$$BO^2 - AO^2 = (a+k)^2 - (a-k)^2 = 4ak$$

34. $(at_1^2, 2at_1) = (2, 4) \Rightarrow t_1 = 1$

$$\text{Now } t_2 = -t_1 - \frac{2}{t_1} = -3;$$

$$\therefore P(2, 4), Q(18, -12)$$

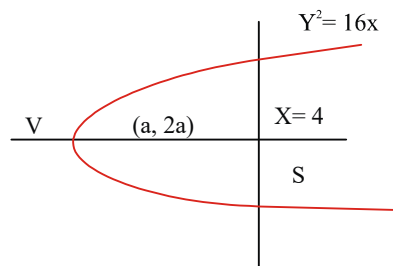
find equation of circle with $\overline{PQ}$ as diameter

35. $(a, 2a)$ is an interior point of $y^2 - 16x = 0$

$$\text{if } (2a)^2 - 16a < 0 \text{ i.e. } a^2 - 4a < 0$$

$V(0,0)$ and $(a, 2a)$ are on the same side of $x-4=0$

So, $a-4 < 0$ i.e. $a < 4$



$$\text{Now } a^2 - 4a < 0 \Rightarrow 0 < a < 4$$

36. It S is the distance of the point (x, y) on the parabola $y = x^2$ from $(0, c)$, then

$$S = \sqrt{x^2 + (y-c)^2} = \sqrt{x^2 + (x^2 - c)^2} \dots(i)$$

Which is least $\Rightarrow S^2$ is least

$$\frac{dS^2}{dx} = 2x + 2(x^2 - c) \times 2x = 0$$

$$\Rightarrow x = 0 \text{ or } x^2 = c - 1/2, c \geq 1/2 \text{ when}$$

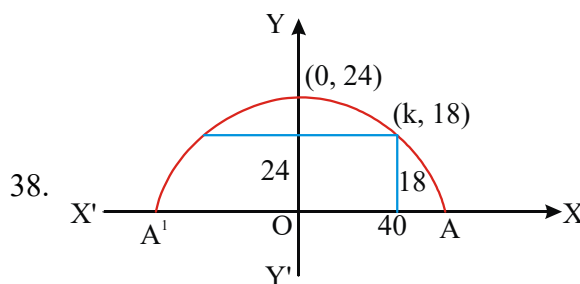
$$x = 0, S = c, \text{ where } x^2 = c - 1/2, c \geq 1/2$$

$$\text{and } S = \sqrt{c - \frac{1}{2} + \frac{1}{4}} = \sqrt{c - \frac{1}{4}}$$

Hence the least distance is

$$c. \text{ if } 0 \leq C \leq 1/2 \text{ and } \sqrt{c - \frac{1}{4}} \text{ if } \frac{1}{2} \leq c \leq 5$$

37. $P(at^2, 2at), S(a, 0)$, Then $Q(-a, 2at)$. Now, $SP = SQ$



$$\text{Vertex } (0, 24) = (h, k)$$

$$\text{parabola } (x-h)^2 = -4a(y-k)$$

$$= x^2 = -4a(y-24) \dots\dots(1)$$

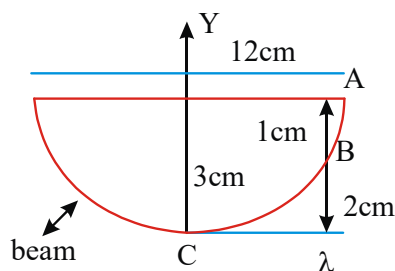
$$\text{passes through } (40, 0) \Rightarrow 4a = \frac{200}{3}$$

from (1) $x^2 = \frac{-200}{3}(y-24)$

$(k, 18) \Rightarrow k^2 = -\frac{200}{3} \times -6 = 400$

$k = 20 \Rightarrow 2k = 40$

39. Equation of Parabola is $x^2 = 4ay$.



point $A\left(6, \frac{3}{100}\right)$ lies on it

$\Rightarrow 6^2 = 4a\left(\frac{3}{100}\right) \Rightarrow a = 300m$

Let AB be the deflection of the beam which is

$\frac{1}{100}m$; $B = \left(x, \frac{2}{100}\right)$ lies on $x^2 = 4(300)y$

$\Rightarrow x^2 = 4 \times 300 \times \frac{2}{100} = 24$; $x = \sqrt{24} = 2\sqrt{6}m$



LEVEL-III

1. Shortest distance between the two parabolas

$y^2 = x - 2, x^2 = y - 2$ is

1) $\frac{1}{4\sqrt{2}}$ 2) $\frac{5}{4\sqrt{2}}$ 3) $\frac{7}{2\sqrt{2}}$ 4) $\frac{6}{7\sqrt{2}}$

2. The name of the conic represented by

$\sqrt{\frac{x}{a}} + \sqrt{\frac{y}{b}} = 1$ is

- 1) circle 2) parabola
3) ellipse 4) hyperbola

3. If the focus of parabola $(y-k)^2 = 4(x-h)$ always lies between the lines $x+y=1$ and $x+y=3$ then

- 1) $0 < h+k < 2$ 2) $0 < h+k < 1$
3) $1 < h+k < 2$ 4) $1 < h+k < 3$

4. Parabola $y^2 = 4a(x-c_1)$ and $x^2 = 4a(y-c_2)$ where c_1 and c_2 are variables, touch each other. Locus of their point of contact is

- 1) $xy = a^2$ 2) $xy = 2a^2$
3) $xy = 4a^2$ 4) $xy = 3a^2$

5. The triangle formed by the tangent to the parabola $y^2 = 4x$ at the point whose abscissa lies in the interval $[a^2, 4a^2]$, the ordinate and the x-axis, has the greatest area equal to

- 1) $12a^3$ 2) $8a^3$ 3) $16a^3$ 4) $20a^3$

6. The straight line $y = mx + c$ ($m > 0$) touches the parabola $y^2 = 8(x+2)$ then the minimum value taken by c is

- 1) 4 2) 8 3) 12 4) 6

7. Minimum distance between the curves $y^2 = 4x$ and $x^2 + y^2 - 12x + 31 = 0$ is

- 1) $\sqrt{5}$ 2) $\sqrt{21}$ 3) $\sqrt{28} - \sqrt{5}$ 4) $\sqrt{21} - \sqrt{5}$

8. Set of values of m for which a chord of slope m of the circle $x^2 + y^2 = 4$ touches parabola $y^2 = 4x$ is

1) $\left(-\infty, -\sqrt{\frac{\sqrt{2}-1}{2}}\right] \cup \left[\sqrt{\frac{\sqrt{2}-1}{2}}, \infty\right)$

2) $(-\infty, -1) \cup (1, \infty)$

3) $(-1, 1)$ 4) $(-\infty, \infty)$

9. Minimum distance between the parabolas $y^2 - 4x - 8y + 40 = 0$ and $x^2 - 8x - 4y + 40 = 0$ is

- 1) 0 2) $\sqrt{3}$ 3) $2\sqrt{2}$ 4) $\sqrt{2}$

10. The Locus of centre of circle which touches given circle externally and the given line is

- 1) Parabola 2) Circle
3) Ellipse 4) Hyperbola

11. A right-angled triangle ABC is inscribed in parabola $y^2 = 4x$, where A is vertex of parabola and $\angle BAC = \pi/2$. If $AB = \sqrt{5}$, then the area of $\triangle ABC$ is

- 1) 20 2) 30 3) 40 4) 50

12. AP is perpendicular to PB, where A is vertex of parabola $y^2 = 4x$ and P on the parabola. B is on the axis of parabola. Then the locus of centroid of ΔPAB is
 1) $9y^2 = 6x + 8$ 2) $9y^2 = 6x - 8$
 3) $9y^2 = -6x - 8$ 4) $9y^2 = -6x + 8$
13. If AB is a focal chord of $x^2 - 2x + y - 2 = 0$ whose focus is 'S'. If $AS = l_1$ then $BS =$
 1) $\frac{l_1}{(l_1 + 1)}$ 2) $\frac{l_1}{(l_1 - 1)}$ 3) $\frac{l_1}{(4l_1 + 1)}$ 4) $\frac{l_1}{(4l_1 - 1)}$
14. A line is drawn from A(-2,0) to intersect the curve $y^2 = 4x$ in P and Q in the first quadrant such that $\frac{1}{AP} + \frac{1}{AQ} < \frac{1}{4}$, then slope of the line is always
 1) $> \sqrt{3}$ 2) $< \frac{1}{\sqrt{3}}$ 3) $> \sqrt{2}$ 4) $> \frac{1}{\sqrt{3}}$
15. A tangent to the parabola $y^2 = 8x$ makes an angle of 45° with the straight line $y = 3x + 5$. Then points of contact are.
 1) $\left(\frac{1}{2}, \frac{1}{2}\right)$ & (8,8) 2) $\left(\frac{1}{2}, -2\right)$ & (8,8)
 3) $\left(\frac{1}{2}, \frac{1}{2}\right)$ & (8,-8) 4) $\left(-\frac{1}{2}, \frac{1}{2}\right)$ & (8,-8)
16. Tangents are drawn from the point (-1,2) to the parabola $y^2 = 4x$. The length that these tangents will intercept on the line $x = 2$ is
 1) $4\sqrt{2}$ 2) $5\sqrt{2}$ 3) $3\sqrt{2}$ 4) $6\sqrt{2}$
17. A parabola of latus-rectum ℓ touches a fixed equal parabola. The axes of two parabolas are parallel. Then the locus of the vertex of the moving parabola is
 1) a parabola whose latus-rectum is 2ℓ
 2) an ellipse 3) a hyperbola
 4) a circle whose radius is 2ℓ
18. An equilateral triangle SAB is inscribed in the parabola $y^2 = 4ax$ having its focus at S. If the chord AB lies towards the left of S then the length of the side of this triangle is
 1. $2a(2 - \sqrt{3})$ 2. $4a(2 - \sqrt{3})$
 3. $a(2 - \sqrt{3})$ 4. $8a(2 - \sqrt{3})$
19. Tangents are drawn from any point on the line $x + 4a = 0$ to the parabola $y^2 = 4ax$. Then the angle subtended by their chord of contact at the vertex is
 1) $\frac{\pi}{6}$ 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{10}$
20. The area of the trapezium whose vertices lie on the parabola $y^2 = 4x$ and its diagonals pass through (1,0) and having length $\frac{25}{4}$ units each is
 1) $25/4$ 2) 25 3) $75/4$ 4) 75
21. Two parabolas have the same focus. If their directrices are the x-axis and the y-axis, respectively, then the slope of their common chord is
 1) ± 1 2) $4/3$ 3) $3/4$ 4) ± 2
22. C is the centre of the circle with centre (0,1) and radius unity. $y = ax^2$ is a parabola. The set of the values of 'a' for which they meet at a point other than the origin, is
 1) $(0, \infty)$ 2) $\left(0, \frac{1}{2}\right)$ 3) $\left(\frac{1}{4}, \frac{1}{2}\right)$ 4) $\left(\frac{1}{2}, \infty\right)$
23. The power of the origin w.r.t the circle on a focal chord of $y^2 = 4ax$ as a diameter is
 1) $-2a^2$ 2) $2a^2$ 3) $3a^2$ 4) $-3a^2$
24. The point $(-2m, m+1)$ is an interior point of the smaller region bounded by the circle $x^2 + y^2 = 4$ and the parabola $y^2 = 4x$. Then m belongs to the interval
 1) $-5 - 2\sqrt{6} < m < 1$ 2) $0 < m < 4$
 3) $-1 < m < \frac{3}{5}$ 4) $-1 < m < -5 + 2\sqrt{6}$
25. PQ is a normal chord of the parabola $y^2 = 4ax$ at $P(at^2, 2at)$. Then the axis of the parabola divides $\overline{PQ}$ in the ratio
 1) $\frac{t^2}{t^2 + 2}$ 2) $\frac{t^2}{t^2 - 2}$ 3) $\frac{2t^2}{t^2 - 2}$ 4) $\frac{t^2}{2(t^2 - 2)}$

26. If the common tangent to the parabola $y^2 = 4ax$ and the circle $x^2 + y^2 = c^2$ makes an angle θ with x axis then $\tan^2 \theta =$
- 1) $-\frac{1}{2} - \frac{\sqrt{c^2 + 4a^2}}{2c}$ 2) $-\frac{1}{2} + \frac{\sqrt{c^2 + 4a^2}}{2c}$
 3) $-\frac{1}{4} + \frac{\sqrt{c^2 + 4a^2}}{2c}$ 4) $-\frac{1}{4} - \frac{\sqrt{c^2 + 4a^2}}{2c}$
27. Radius of the largest circle which passes through the focus of the parabola $y^2 = 4x$ and contained in it, is
- 1) 8 2) 4 3) 2 4) 5
28. The locus of point of intersection of the two tangents to the parabola $y^2 = 4ax$ which intercept a given distance $4c$ on the tangent at the vertex is
- 1) $y^2 - 4ax = \frac{c^2}{4}$ 2) $y^2 - 4ax = 8c^2$
 3) $y^2 - 4ax = \frac{c^2}{2}$ 4) $y^2 - 4ax = 16c^2$
29. Through the vertex A of the parabola $y^2 = 4ax$ chords AP and AQ are drawn at right angles to one another. Then the line PQ meets the axis in a fixed point with coordinates
- 1) $(3a, 0)$ 2) $(2a, 0)$ 3) $(6a, 0)$ 4) $(4a, 0)$
30. Through the vertex O of the parabola $y^2 = 4ax$ chords OP, OQ are drawn at right angles to each other. If the locus of mid points of the chord PQ is a parabola then its vertex is
- 1) $(4a, 0)$ 2) $(2a, 0)$ 3) $(a, 0)$ 4) $(3a, 0)$
31. A movable parabola touches the x axis and the y-axis at $(1, 0)$ and $(0, 1)$. Then the locus of the focus of the parabola is
- 1) $2x^2 - 2x + 2y^2 - 2y + 1 = 0$
 2) $x^2 - 2x + 2y^2 - 2y + 1 = 0$
 3) $2x^2 - 2x + 2y^2 + 2y + 2 = 0$
 4) $2x^2 + 2x - 2y^2 - 2y - 2 = 0$
32. A variable chord PQ of the parabola $y^2 = 4ax$ is drawn parallel to $y=x$. Then the locus of point of intersection of normals at P and Q is
- 1) $2x - y - 12a = 0$ 2) $2x - y + 10a = 0$
 3) $2x - y - 8a = 0$ 4) $2x - y + 6a = 0$
33. The locus of point of intersection of two normals drawn to the parabola $y^2 = 4ax$ which are at right angles is
- 1) $y^2 = a(x - 3a)$ 2) $y^2 = a(x - a)$
 3) $y^2 = 3a(x - 2a)$ 4) $y^2 = 2a(x - 2a)$
34. If the normal to the parabola $y^2 = 4ax$ at the point P $(at^2, 2at)$ cuts the parabola again at Q $(aT^2, 2aT)$ then
- 1) $-2 \leq T \leq 2$ 2) $T \in (-\infty, -8) \cup (8, \infty)$
 3) $T^2 < 8$ 4) $T^2 \geq 8$
35. Consider a circle with its centre lying on the focus of the parabola $y^2 = 2px$ such that it touches the directrix of the parabola. Then a point of intersection of the circle and the parabola is
- 1) $\left(\frac{p}{2}, p\right)$ or $\left(\frac{p}{2}, -p\right)$ 2) $\left(\frac{p}{2}, \frac{p}{2}\right)$
 3) $\left(-\frac{p}{2}, p\right)$ 4) $\left(-\frac{p}{2}, -\frac{p}{2}\right)$
36. Range of values of k for which the point $(k, -1)$ is exterior to both the parabolas $y^2 = |x|$ is
- 1) $(-1, 0)$ 2) $(-1, 1)$ 3) $(0, 1)$ 4) $(0, -1)$
37. If S is the focus of the parabola $y^2 = 8x$. P is a point on the parabola. The normal at P meets the axis in G. If SPG is an equilateral triangle then P is
- 1) $(6, 4\sqrt{3})$ 2) $(8, 8)$ 3) $(4, 4\sqrt{2})$ 4) $(3, 2\sqrt{3})$
38. A point on the parabola $y^2 = 18x$ at which the ordinate increases at twice the rate of the abscissa is
- 1) $(2, 4)$ 2) $(2, -4)$ 3) $\left(-\frac{9}{8}, \frac{9}{2}\right)$ 4) $\left(\frac{9}{8}, \frac{9}{2}\right)$
39. Let P be the point $(1, 0)$ and Q be a point on the parabola $y^2 = 8x$. The locus of midpoint PQ is
- 1) $x^2 + 4y + 2 = 0$ 2) $x^2 - 4y + 2 = 0$
 3) $y^2 - 4x + 2 = 0$ 4) $y^2 + 4x + 2 = 0$

40. The locus of the vertices of the family of parabolas $y = \frac{a^2 x^2}{3} + \frac{a^2 x}{2} - 2a$ is
 1) $xy = \frac{35}{16}$ 2) $xy = \frac{34}{105}$ 3) $xy = \frac{105}{64}$ 4) $xy = \frac{3}{4}$
41. The circle $x^2 + y^2 + 2\lambda x = 0, \lambda \in R$ touches the parabola $y^2 = 4x$ externally, then
 1) $\lambda > 0$ 2) $\lambda < 0$ 3) $\lambda > 1$ 4) $\lambda = 0$
42. A circle, with its centre at the focus of the parabola $y^2 = 4ax$, & touching its directrix and the parabola at the points A, B, then length AB equal to
 1) $2a$ 2) $3a$ 3) a 4) $4a$
43. If normal is drawn to $y^2 = 12x$ making an angle 45° with the axis then the foot of the normal is
 1) $(3, 8)$ 2) $(3, -6)$ 3) $(12, -12)$ 4) $(8, -8)$
44. If from a point A, the two tangents drawn to the parabola $y^2 = 4ax$ are normals to the parabola, $x^2 = 4by$ then
 1) $a^2 > 2b^2$ 2) $a^2 < 4b^2$ 3) $a^2 > 8b^2$ 4) $a^2 = 7b^2$
45. If $\left(\frac{a}{b}\right)^{1/3} + \left(\frac{b}{a}\right)^{1/3} = \frac{\sqrt{3}}{2}$, then the angle of intersection of the parabola $s y^2 = 4ax$ and $x^2 = 4by$ at the point other than the origin is
 1) $\frac{\pi}{4}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{2}$ 4) $\frac{\pi}{6}$
46. A variable chord PQ of the parabola $y^2 = 4ax$ is drawn parallel to $y = x$, then the locus of point of intersection of normals at P and Q is
 1) $2x - y - 12a = 0$ 2) $2x - y - 10a = 0$
 3) $2x - y - 8a = 0$ 4) $2x - y + 6a = 0$
47. The locus of point of intersection of two normals drawn to the parabola $y^2 = 4ax$ which are at right angle is
 1) $y^2 = a(x - 3a)$ 2) $y^2 = a(x - a)$
 3) $y^2 = 3a(x - 2a)$ 4) $y^2 = 2a(x - 2a)$
48. Let $y = 4x^2$ & $\frac{x^2}{a^2} - \frac{y^2}{16} = 1$ intersect if
 1) $|a| \leq \frac{1}{\sqrt{2}}$ 2) $a \leq -\frac{1}{\sqrt{2}}$ 3) $a > -\frac{1}{\sqrt{2}}$ 4) $|a| \geq -\frac{1}{\sqrt{2}}$
49. The equation of the common tangent touching the circle $(x - 3)^2 + y^2 = 9$ and the parabola $y^2 = 4ax$ above the x-axis is
 1) $\sqrt{3}y = 3x + 1$ 2) $\sqrt{3}y = -(x + 3)$
 3) $\sqrt{3}y = (x + 3)$ 4) $\sqrt{3}y = -(3x + 1)$
50. The equation of the common tangent to the parabola $y = x^2$ and $y = -(x - 2)^2$ is
 1) $y - 4(x - 1)$ 2) $x = 0$
 3) $y = -4(x - 1)$ 4) $y = 30x - 50$
51. A line bisecting the ordinate PN of a point $P(at^2, 2at), t > 0$, on the parabola $y^2 = 4ax$ is drawn parallel to the axis to meet the curve at Q. If NQ meets the tangents at the vertex at the point T. Then the coordinates of T are.
 1) $\left(0, \frac{4}{3}at\right)$ 2) $(0, 2at)$ 3) $\left(\frac{1}{4}at^2, at\right)$ 4) $(0, at)$
52. If the line $x - 1 = 0$ is the directrix of the parabola $y^2 - kx + 8 = 0, k \neq 0$ and the parabola intersects the circle $x^2 + y^2 = 4$ in two real distinct points, then the value of k is
 1) -4 2) -8 3) 4 4) 8
53. Over the towers of a bridge a cable is hung in the form of a parabola, have their tops 30 metres above the road-way are 200 metres apart. If the cable is 5 metres above the road way at the centre of the bridge, then the length of the vertical supporting cable 30 metres from the centre is
 1) 100 m 2) 30 m 3) $9/4$ m 4) $29/4$ m
54. The vertex of a parabola is the point (a, b) and latusrectum is of length l . If the axis of the parabola is along the positive direction of y-axis, then its equation is
 1) $(x + a)^2 = \frac{l}{2}(2y - 2b)$ 2) $(x - a)^2 = \frac{l}{2}(2y - 2b)$
 3) $(x + a)^2 = \frac{l}{4}(2y - 2b)$ 4) $(x - a)^2 = \frac{l}{8}(2y - 2b)$

LEVEL - III - KEY

- 1) 3 2) 2 3) 1 4) 3 5) 3 6) 1 7) 1
 8) 1 9) 4 10) 1 11) 1 12) 2 13) 4 14) 1
 15) 2 16) 4 17) 1 18) 2 19) 2 20) 3 21) 1
 22) 4 23) 4 24) 4 25) 1 26) 2 27) 2 28) 4
 29) 4 30) 1 31) 1 32) 1 33) 1 34) 4 35) 1
 36) 2 37) 1 38) 4 39) 3 40) 3 41) 1 42) 4
 43) 2 44) 3 45) 2 46) 1 47) 1 48) 1 49) 3
 50) 1 51) 1 52) 2 53) 4 54) 2

LEVEL-III - HINTS

1. Any point on $y^2 = x - 2$ is $(t^2 + 2, t)$
Curves are symmetrical about the line $y = x$

$$\text{distance } D = \frac{t^2 + 2 - t}{\sqrt{2}} \Rightarrow \frac{dD}{dt} = 0$$

$$\Rightarrow \frac{1}{\sqrt{2}}(2t - 1) = 0 \Rightarrow t = \frac{1}{2}$$

Required points are $\left(\frac{9}{4}, \frac{1}{2}\right)$ and $\left(\frac{1}{2}, \frac{9}{4}\right)$

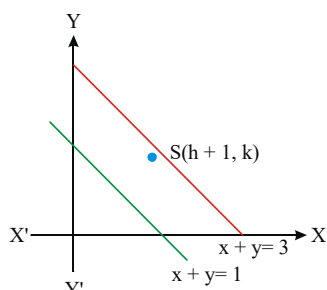
Then find the distance between the above points.

$$2. \quad \sqrt{\frac{x}{a}} + \sqrt{\frac{y}{b}} = 1 \quad ; \quad \sqrt{\frac{x}{a}} = 1 - \sqrt{\frac{y}{b}} \quad \text{squaring}$$

$$\frac{x}{a} = 1 + \frac{y}{b} - 2\sqrt{\frac{y}{b}} \quad ; \quad \frac{y}{b} - \frac{x}{a} + 1 = 2\sqrt{\frac{y}{b}} \quad \text{again squaring then apply the conditions}$$

$$H^2 = AB \text{ \& } \Delta \neq 0$$

3. Coordinate of focus will be $(h+1, k)$ Now focus should lie to the opposite side of origin with respect to with respect to line. $x + y - 1 = 0$ and same side as origin with respect to line $x + y - 3 = 0$. Hence $h + k > 0$ and $h + k < 2$.



4. Let $P(x, y)$ be the point of contact
 $\therefore 2y \frac{dy}{dx} = 4a$ and $2x = 4a \frac{dy}{dx}$ For the tangency of curves, $\frac{4a}{2y} = \frac{2x}{4a} \Rightarrow xy = 4a^2$, which is the required locus

5. Let $P = (h^2, 2h)$. Slope of tangent at $P = \tan \theta = \frac{1}{h}$

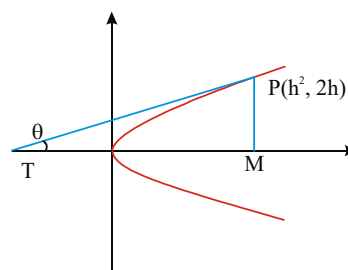
$$\text{Thus the area of } \Delta PTM = \frac{1}{2} \times PM \times TM =$$

$$\frac{1}{2} \times 2h \times 2h \cot \theta = 2h^3$$

The area will be maximum when h is maximum

$$\therefore a^2 \leq h^2 \leq 4a^2 \text{ for maximum } h, h = 2a$$

$$\therefore \text{Maximum area} = a(2a)^3 = 16a^3$$



6. Tangent to $y^2 = 8(x+2)$ is $y = m(x+2) + \frac{2}{m}$

$$c = 2m + \frac{2}{m} \Rightarrow \frac{c}{2} = \left(m + \frac{1}{m}\right)$$

$$\therefore m + \frac{1}{m} \geq 2 \Rightarrow \frac{c}{2} \geq 2 \Rightarrow c \geq 4 \Rightarrow$$

The minimum value of $c = 4$.

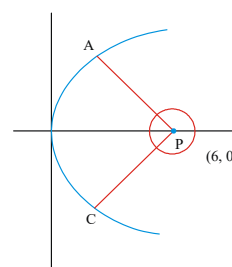
7. Centre and radius of the given circle is $P(6, 0)$ and $\sqrt{5}$ respectively. Equation of normal for $y^2 = 4x$ at $(t^2, 2t)$ is $y = -tx + 2t + t^3$, it must pass through $(6, 0)$ in order that it gives minimum distance between the two curves.

$$\therefore 0 = t^3 - 4t \Rightarrow t = 0 \text{ or } t = \pm 2$$

$$\therefore A(4, 4) \text{ and } C(4, -4)$$

$$PA = PC = \sqrt{20} = 2\sqrt{5}$$

$$\therefore \text{required minimum distance} = 2\sqrt{5} - \sqrt{5} = \sqrt{5}.$$



8. Equation of tangent to the parabola with slope m is $y = mx + \frac{1}{m}$ For this line to be chord of the circle

$$x^2 + y^2 = 4, \text{ where } \left| \frac{\frac{1}{m}}{\sqrt{1+m^2}} \right| < 2 \therefore 1 < 4m^2(1+m^2)$$

$$4m^4 + 4m^2 - 1 > 0 \Rightarrow m^2 > \frac{-1 + \sqrt{2}}{2}$$

$$\Rightarrow m \in \left(-\infty, -\sqrt{\frac{-1 + \sqrt{2}}{2}} \right) \cup \left(\sqrt{\frac{-1 + \sqrt{2}}{2}}, \infty \right)$$

9. Since two parabolas are symmetrical about $y = x$
So, minimum distance is zero if they intersect
 $y = x$ on solving $y = x$ & $y^2 - 4x - 8y + 4 = 0$
we get $x^2 - 12x + 40 = 0$ which has no real solution so they do not intersect.

Minimum distance is between tangents to the parabola parallel to $y = x$

Differentiating $x^2 - 8x - 4y + 40 = 0$ with respect to x , we get

$$2x - 8 - 4y' = 0 \quad y' = \frac{x - 4}{2} = 1 \text{ (slope of tangent)}$$

so, $x = 6$ and $y = 7$

Point on parabola $(x - 4)^2 = 4(y - 6)$ is $(6, 7)$
and corresponding point on

$$(y - 4)^2 = 4(x - 6) \text{ is } (7, 6)$$

So, minimum distance $= \sqrt{2}$.

10. Given circle is $x^2 + y^2 = a^2$. Let another circle with centre (h, k) and touches the line $x = b$

$$\Rightarrow c_1 c_2 = r_1 + r_2 \Rightarrow \sqrt{h^2 + k^2} = a + (b - h) = (a + b) - h$$

square on both sides

$$h^2 + k^2 = (a + b)^2 + h^2 - 2h(a + b)$$

$$\Rightarrow \text{Locus of } (h, k) \text{ is } y^2 = (a + b)^2 - 2x(a + b)$$

Which is equation of parabola

11. $A = (0, 0), B = (T_1^2, 2t_1); C = (T_2^2, 2t_2)$

$$AB = \sqrt{5} \Rightarrow t_1 = \pm 1; \Rightarrow B = (1, 2)$$

$$AB \perp AC \Rightarrow t_2 = -4; \Rightarrow C = (16, -8) \therefore \text{Area} = 20$$

12. $A = (0, 0); P(t^2, 2t)$

Find equation of BP and hence x -intercept

$$B = (t^2 + 4, 0). \text{ Find locus of centroid of } \triangle APB$$

$$13. \frac{1}{SA} + \frac{1}{SB} = \frac{2}{l}$$

14. Apply parametric equation of a line

$$\frac{1}{r_1} + \frac{1}{r_2} < \frac{1}{4} \Rightarrow \cos \theta < \frac{1}{2} \Rightarrow \theta > 60 \Rightarrow m > \sqrt{3}$$

$$15. m = 3; \alpha = 45^\circ \text{ Slope} = \frac{m \pm \tan \alpha}{1 \mp m \tan \alpha} \Rightarrow m = -2, \frac{1}{2}$$

$$\text{Use } \left(\frac{a}{m^2}, \frac{2a}{m} \right)$$

16. Find $S_1^2 = S \cdot S_{11}$; Solve with $x = 2$ to get quadratic in y . Now find $|y_1 - y_2|$

17. Let the fixed parabola $y^2 = 4ax \dots (1)$

moving parabola is $(y - k)^2 = -4a(x - h) \dots (2)$
on solving (1), (2) we get

$$(y - k)^2 = -4a \left(\frac{y^2}{4a} - h \right)$$

Then apply the discriminant of the quadratic equation is '0'. $\Rightarrow 4k^2 - 8(k^2 - 4ah) = 0$

$$\Rightarrow -4k^2 + 32ah = 0$$

$$\Rightarrow k^2 = 8ah \text{ the locus of the vertex is } y^2 = 8ax$$

whose latusrectum is $8a = 2(4a) = 2l$

18. $A = (at_1^2, 2at_1), B = (at_2^2, -2at_1)$

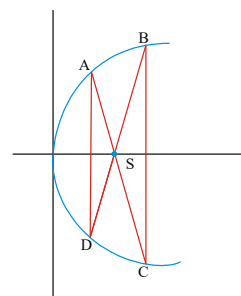
The slope of AS is $\tan \frac{5\pi}{6}$

$$\frac{2at_1}{at_1^2 - a} = -\frac{1}{\sqrt{3}} \Rightarrow t_1^2 + 2\sqrt{3}t_1 - 1 = 0$$

$$\Rightarrow t_1 = -\sqrt{3} \pm 2; t_1 = -\sqrt{3} - 2 \text{ rejected}$$

$$\therefore t_1 = 2 - \sqrt{3} \text{ then } AB = 4at_1 = 4a(2 - \sqrt{3})$$

19. $(at_1, t_2, a(t_1 + t_2))$ lies on $x = -4a \Rightarrow t_1 t_2 = -4$
Now find angle subtended at the vertex



- 20.

$$A = (t^2, 2t); S = (1, 0); AC = BD = \frac{25}{4}$$

$$\text{Focal distance } AS = t^2 + 1; CS = \frac{21 - 4t^2}{4}$$

$$\frac{1}{CS} + \frac{1}{AS} = \frac{1}{a} \Rightarrow t = \pm 2, \pm 1/2$$

which are the points. Now find area.

21. Let focus = (a, b) Parabola equations are

$$(x-a)^2 + (y-b)^2 = x^2;$$

$$(x-a)^2 + (y-b)^2 = y^2$$

$$\text{Subtract} \Rightarrow x^2 - y^2 = 0 \Rightarrow \text{Slopes} = \pm 1$$

22. Solve $x^2 = \frac{y}{a}, x^2 + (y-1)^2 = 1$

$$\text{We get } y = 2 - \frac{1}{a} \text{ Put in } y = ax^2$$

$$x^2 = \frac{2a-1}{a^2} > 0 \Rightarrow a > \frac{1}{2}$$

23. Extrimities of the focal chord are $(at^2, 2at)$ and

$$\left(\frac{a}{t^2}, \frac{-2a}{t}\right).$$

equation of circle with focal chord as diameter

$$(x-at^2)\left(x-\frac{a}{t^2}\right) + (y-2at)\left(y+\frac{2a}{t}\right) = 0$$

find power of origin $(0,0)$ w.r.t the above circle.

24. $(-2m, m+1)$ lies inside the circle and parabola

$$\text{then } (-2m)^2 + (m+1)^2 - 4 < 0 \text{ and}$$

$$(m+1)^2 - 4(-2m) < 0 \Rightarrow 5m^2 + 2m - 3 < 0$$

$$\text{and } m^2 + 10m + 1 < 0 \Rightarrow -1 < m < \frac{3}{5} \dots (1) \text{ and}$$

$$-5 - 2\sqrt{6} < m < -5 + 2\sqrt{6} \dots (2)$$

$$\text{from (1) and (2) } -1 < m < -5 + 2\sqrt{6}$$

25. $P(at^2, 2at), Q(at_2^2, 2at_2)$ then $t_2 = -t - \frac{2}{t}$

X-axis divides in the ratio $-y_1 : y_2$

26. $y = mx + \frac{a}{m}$ is a tangent to $y^2 = 4ax$

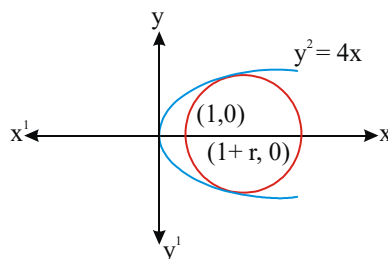
$$\text{It is tangent to } x^2 + y^2 = c^2 \Rightarrow c = \frac{a}{m\sqrt{1+m^2}}$$

$$\Rightarrow c^2 m^4 + c^2 m^2 - a^2 = 0 \therefore m^2 = \frac{-c^2 + \sqrt{c^4 + 4a^2 c^2}}{2c^2}$$

$$\Rightarrow \tan^2 \theta = -\frac{1}{2} + \frac{\sqrt{4a^2 + c^2}}{2c}$$

27. Let r be the radius of the largest circle passing through the focus $(1,0)$ of $y^2 = 4x$.

Clearly, centre of the circle will be on x-axis its co-ordinates are $(1+r, 0)$



The equation of the circle is $(x-1-r)^2 + y^2 = r^2$

It touches $y^2 = 4ax$. Therefore, the equation

$$(x-r-1)^2 + 4x = r^2 \text{ must have equal roots.}$$

$$\Rightarrow 4(1-r)^2 - 4(2r+1) = 0 \Rightarrow r = 4$$

28. Standard result $y^2 - 4ax = (4c)^2$

$$29. m_1 m_2 = -1 \Rightarrow t_1 t_2 = -4$$

$$\text{Equation of } \overline{PQ} \text{ is } y(t_1 + t_2) - 2x - 2at_1 t_2 = 0$$

$$y = 0 \Rightarrow x = 4a$$

30. Let midpoint of $\overline{PQ}$ is $P(x_1, y_1) \therefore \overline{PQ}$ is $S_1 = S_{11}$
Now, homogenise the given curve

31. since, x -axis and y -axis are two perpendicular tangents to the parabola and both meet at origin then directrix passes through the origin. Let $y = mx$ be the directrix $S(h, k)$ is the focus

$$\Rightarrow \sqrt{(h-1)^2 + k^2} = \left| \frac{m(1)-0}{\sqrt{m^2+1}} \right| \dots (1)$$

$$\text{and } \sqrt{(h-0)^2 + (k-1)^2} = \frac{1}{\sqrt{1+m^2}} \dots (2)$$

from (1) and (2) squaring and adding $(h-1)^2 + k^2 + h^2 + (k-1)^2 = 1$ expanding the locus of (h, k) is $2x^2 - 2x + 2y^2 - 2y + 1 = 0$

32. Let $P(at_1^2, 2at_1), Q(at_2^2, 2at_2)$

$$\text{slope of } \overline{PQ} \text{ is } 1 \Rightarrow t_1 + t_2 = 2 \text{ then}$$

$$(x, y) = (2a + a((t_1 + t_2)^2 - t_1 t_2), -at_1 t_2(t_1 + t_2))$$

33. P.I of normals at t_1, t_2 is

$$x = 2a + a((t_1 + t_2)^2 - t_1 t_2); y = -at_1 t_2(t_1 + t_2)$$

$$\text{given } t_1 t_2 = -1$$

34. Let PQ is normal at P and is also a chord PQ . Comparing their equations, we get

$$T = -t - \frac{2}{t} \text{ or } t^2 - tT + 2 = 0$$

Since, both t and T are real

$$\therefore D \geq 0 \therefore T^2 - 8 \geq 0 \text{ or } T^2 \geq 8$$

35. The equation of circle will be

$$\left(x - \frac{p}{2}\right)^2 + y^2 = (p)^2 \Rightarrow x^2 + y^2 - px - \frac{3p^2}{4} = 0$$

Solving it with $y^2 = 2px$, we get

$$x^2 + px - \frac{3p^2}{4} = 0 \Rightarrow x = -\frac{3p}{2} \text{ or } \frac{p}{2},$$

$$x > 0 \Rightarrow \frac{p}{2} \text{ and then } y = \pm p$$

36. The two parabolas are $y^2 = x$ and $y^2 = -x$. The point $(k, -1)$ is an exterior point.

$$\text{if } (-1)^2 - k > 0 \text{ and } (-1)^2 + k > 0$$

$$\text{if } 1 - k > 0 \text{ and } 1 + k > 0$$

$$\text{if } k < 1 \text{ and } k > -1$$

$$\text{if } -1 < k < 1 \text{ Hence Range of } k \text{ is } (1, 1)$$

37. Let $P(at^2, 2at)$, $G(2a + at^2, 0)$ and $S(a, 0)$

$$SP = PG \Rightarrow t^2 = 3$$

$$38. \frac{dy}{dx} = 2 \Rightarrow \frac{18}{2y} = 2 \Rightarrow y = \frac{9}{2}$$

$$\text{Putting in } y^2 = 18x \text{ we get } x = \frac{9}{8}$$

$$\text{so point is } \left(\frac{9}{8}, \frac{9}{2}\right).$$

39. $P = (1, 0)Q(h, k)$ such that $k^2 = 8h$

Let (α, β) be mid point of PQ

$$\therefore \alpha = \frac{h+1}{2}, \beta = \frac{k+0}{2}, 2\alpha - 1 = h, 2\beta = k$$

$$\therefore (2\beta)^2 = 8(2\alpha - 1) \Rightarrow \beta^2 = 4\alpha - 2$$

$$\Rightarrow y^2 - 4x + 2 = 0$$

$$40. y = \frac{a^2 x^2}{3} + \frac{a^2 x}{2} - 2a = \frac{a^2}{3} \left[x^2 + \frac{3}{2a} x \right] - 2a$$

$$= \frac{a^2}{3} \left[x + \frac{3}{4a} \right]^2 - \frac{9}{16a^2} X \frac{a^3}{3} - 2a$$

$$\Rightarrow y + \frac{3a}{16} + 2a = \frac{a^3}{3} \left(x + \frac{3}{4a} \right)^2$$

$$\Rightarrow y + \frac{35}{16} a = \frac{a^3}{3} \left(x + \frac{3}{4a} \right)^2$$

$$\therefore \text{Vertex (h,k) is } \left(\frac{-3}{4a}, \frac{-35a}{16}\right)$$

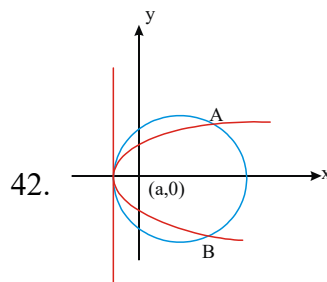
$$\Rightarrow h = \frac{-3}{4a}, k = \frac{-35a}{16} \Rightarrow hk = \frac{105}{64}$$

$$\therefore \text{Locus is } xy = \frac{105}{64}$$

41. Given equation are $x^2 + y^2 + 2\lambda x = 0$. $y^2 = 4x$

centre of circle is $(-\lambda, 0)$ and radius is $\sqrt{\lambda^2 + 0} = \lambda$.

Clearly the circle will touch the parabola externally if centre of circle lies is -ve x-axis.



42.

43. Foot of normal $(am^2, -2am)$

44. Any point on the parabola $x^2 = 4by$ is $(2bt, bt^2)$.

$$\therefore \frac{dy}{dx} = \frac{x}{2b}$$

$$\therefore \text{Slope of normal at } (2bt, bt^2) \text{ is } \frac{-2b}{2bt} = -\frac{1}{t}$$

$$\text{Equation of normal is } y - bt^2 = -\frac{1}{t}(x - 2bt)$$

$$-ty - bt^3 = -x + 2bt \Rightarrow y = \frac{x}{t} + (2b + bt^2)$$

Now, equation (1) is tangent to the parabola

$$\therefore 2b + bt^2 = \frac{a}{-\frac{1}{t}} = -at$$

$$y^2 = 4ax$$

$$\{\text{Condition of Tangency is } c = \frac{a}{m}\} \Rightarrow bt^2 + at + 2b = 0$$

For distinct real roots, discriminant > 0

$$\Rightarrow a^2 - 8b^2 > 0 \therefore a^2 > 8b^2$$

45. Find the slopes of both the curves m_1 & m_2 and

$$\text{use the formula } \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

46. Let $P(at_1^2, 2at_1)$, $Q(at_2^2, 2at_2)$

Slope of $\overline{PQ}$ is $1 \Rightarrow t_1 + t_2 = 2$ then apply the formula of point of intersection of normals at parametric point.

47. P.I of normals at t_1, t_2 is $x = 2a + a((t_1 + t_2)^2 - t_1 t_2)$

$$y = -at_1 t_2 (t_1 + t_2) \text{ given } t_1 t_2 = -1$$

$$48. \quad y = 4x^2 \text{ \& } \frac{1}{4}y = x^2 \text{ using } \frac{1}{4a^2}y - \frac{y^2}{16} = 1$$

$$\Rightarrow 4y - a^2y^2 = 16a^2 \Rightarrow a^2y^2 - 4y + 16a^2 = 0$$

$$\Rightarrow D \geq 0 \text{ for intersection of two curves}$$

$$\Rightarrow 16 - 4a^2(16a^2) \geq 0 \Rightarrow 1 - 4a^4 \geq 0$$

$$\Rightarrow (2a^2)^2 \leq 1 \Rightarrow -\frac{1}{\sqrt{2}} \leq a \leq \frac{1}{\sqrt{2}}$$

$$49. \quad \text{Let the common tangent to the parabola } y^2 = 4x$$

be $y = mx + \frac{1}{m}$. It should also touch the circle $(x-3)^2 + y^2 = 9$ whose centre is at $(3,0)$ and radius = 3, then

$$\frac{\left|3m + \frac{1}{m}\right|}{\sqrt{1+m^2}} = 3 \Rightarrow 9m^4 + 1 + 6m^2 =$$

$$= 9m^2(1+m^2) \Rightarrow 3m^2 = 1 \Rightarrow m = \pm \frac{1}{\sqrt{3}}$$

But $m > 0$ then equation of common tangent is

$$y = \frac{1}{\sqrt{3}}x + \sqrt{3} \text{ or } \sqrt{3}y = x + 3$$

$$50. \quad \text{If } y = mx + c \text{ is tangent to } y = x^2 \text{ then}$$

$$x^2 - mx - c = 0 \text{ has equal roots}$$

$$\Rightarrow m^2 + 4c = 0 \Rightarrow c = -\frac{m^2}{4}$$

$$\therefore y = mx - \frac{m^2}{4} \text{ is a tangent to } y = x^2$$

$$\therefore \text{This is also tangent to } y = -(x-2)^2$$

$$\Rightarrow mx - \frac{m^2}{4} = -x^2 + 4x - 4$$

$$\Rightarrow x^2 + (m-4)x + \left(4 - \frac{m^2}{4}\right) = 0 \text{ has equal roots.}$$

$$m^2 - 8m + 16 = -m^2 + 16 \Rightarrow m = 0, 4$$

$$\therefore y = 0 \text{ or } y = 4x - 4 \text{ are the tangents.}$$

$$51. \quad \text{Equation of the line parallel to the axis and bisecting the ordinate PN of the point } P(at^2, 2at) \text{ is}$$

$$y = at \text{ which meet the parabola } y^2 = 4ax \text{ at the}$$

point $Q\left\{\left(\frac{1}{4}\right)at^2, at\right\}$. Coordinates of N are $(at^2, 0)$.

Equation of NQ is $y = \frac{0-at}{at^2 - \left(\frac{1}{4}\right)at^2}(x - at^2)$

Which meets the tangent at the vertex, $x = 0$ at the point $y = (4/3)at$

$$52. \quad \text{The equation of the parabola can be written as}$$

$$y^2 = k(x - 8/k) \text{ which is of the form } Y^2 = 4AX$$

Where $Y = y, X = x - 8/k$ and $A = K/4$. Equation of the directrix is $X = -A \Rightarrow x - 8/k = -k/4$ which represents the given line $x - 1 = 0$ if $\frac{8}{k} - \frac{k}{4} = 1$

$$\Rightarrow k^2 + 4k - 32 = 0 \Rightarrow k = -8 \text{ or } 4$$

For $k = 4$, the parabola is $y^2 = 4(x - 2)$ whose vertex is $(2,0)$ and touches the circle $x^2 + y^2 = 4$ at the vertex. Therefore $x^2 + y^2 = 4$. For $k = -8$, the parabola is $y^2 = -8(x + 1)$. Which intersects the circle $x^2 + y^2 = 4$ at two real distinct points as the vertex $(-1,0)$ of the parabola lies inside the circle.

$$53. \quad \text{Suppose the bridge is hung in a parabolic arc with the vertex at the lowest point and the axis vertical. Then the equation of the parabola takes the form}$$

$$x^2 = 4ay. \text{ Since it passes through the point } (100, 25), \text{ we have } (100)^2 = 4a(25) \text{ or}$$

$$a = \frac{100 \times 100}{25 \times 4} = 100. \text{ The length of the specified vertical supporting cable is given by } y + 5, \text{ where } y \text{ is the ordinate of the point } P(30, y) \text{ on the parabola}$$

$$x^2 = 400y \text{ whose abscissa is } 30, \text{ thus}$$

$$(30)^2 = 400y \text{ or } y = \frac{30 \times 30}{400} = \frac{9}{4}$$

Therefore, the required length is given by

$$PQ = y + 5 = \frac{9}{4} + 5 = \frac{29}{4} = 7\frac{1}{4}m$$

$$54. \quad \text{Let the equation of parabola along positive direction of y-axis is } (x-a)^2 = 4\alpha(y-b) \text{ it passes through } (a,b)$$

Length of latus rectum is $4\alpha = l$

$$\therefore \text{equation (1) reduces to } (x-a)^2 = l(y-b).$$



LEVEL-IV

STATEMENT TYPE

Statement-1 (Assertion) and Statement-2 (Reason). Each of these question also has four alternative choices, only one of which is the correct answer. You have to select the correct choice.

- 1) Statement-1 is true, Statement-2 is true, Statement-2 is the correct explanation for statement-1
- 2) Statement-1 is true, Statement-2 is true, statement-2 is not the correct explanation for statement-1.
- 3) Statement-1 is true, Statement-2 is false.
- 4) Statement-1 is false, statement-2 is true.

1. **Statement-1** : If $x + y = k$ is a normal to the parabola $y^2 = 12x$ then k is 9.

Statement-2 : Equation of normal to the parabola $y^2 = 4ax$ is $y - mx + 2am + am^3 = 0$

2. **Statement-1** : The normals at the point $(4,4)$ and $\left(\frac{1}{4}, -1\right)$ of the parabola $y^2 = 4x$ are perpendicular.

Statement - 2 : The tangents to the parabola at the end of a focal chord are perpendicular.

3. **Statement - 1** : The point $(\sin \alpha, \cos \alpha)$ does not lie outside the parabola $y^2 + x - 2 = 0$ when $\alpha \in \left[\frac{\pi}{2}, \frac{5\pi}{6}\right] \cup \left[\pi, \frac{3\pi}{2}\right]$.

Statement-2 : The point (x_1, y_1) lies outside the parabola $y^2 = 4ax$ if $y_1^2 - 4ax_1 > 0$.

4. **Statement-1** : The curve $y = \frac{-x^2}{2} + x + 1$ is symmetric with respect to the line $x=1$.

Statement 2 : A parabola is symmetric about its axis

5. **Statement - 1** : Number of focal chords of length 6 units that can be drawn on the parabola $y^2 - 2y - 8x + 17 = 0$ is zero.

Statement - 2 : Latus rectum is the shortest focal chord of the parabola.

6. **Statement - 1** : A is a point on the parabola $y^2 = 4ax$. The normal at A cuts the parabola again at point B. If AB subtends a right angle at the vertex of the parabola, then slope of AB is $\sqrt{2}$.

Statement-2 : If normal at $(at_1^2, 2at_1)$ cuts again the parabola at $(at_2^2, 2at_2)$ then $t_2 = t_1 - \frac{2}{t_1}$.

7. **Statement-1** : Two parabolas $y^2 = 4ax$ and $x^2 = 4ay$ have common tangent $x + y + a = 0$.

Statement - 2 : $x + y + a = 0$ is common tangent to the parabolas $y^2 = 4ax$ and $x^2 = 4ay$ and point of contact lie on their respective end points of latus rectum.

8. **Statement - 1** : Normal chord drawn at the point $(8,8)$ of the parabola $y^2 = 8x$ subtends a right angle at the vertex of the parabola.

Statement - 2 : Every chord of the parabola $y^2 = 4ax$ passing through the point $(4a, 0)$ subtends a right angle at the vertex of the parabola.

MATCH THE FOLLOWING

S.NO	COLUMN-I	COLUMN-II
A)	Radius of the largest circle which passes through the focus of the parabola $y^2 = 4x$ and contained in it is	(p) 16
B)	Two perpendicular tangents PA & PB are drawn to the parabola $y^2 = 16x$ the min AB is	(q) 5
C)	The shortest distance between the parabola $y^2 = 4x$ and $y^2 = 2x - 6$ is 'd' then $d^2 =$	(r) 8
D)	The harmonic mean of the segments of a focal chord of the parabola $y^2 = 8x$	(s) 4

- 1) A-r, B-s, C-p, D-q 2) A-s, B-r, C-q, D-p
3) A-p, B-s, C-q, D-r 4) A-r, B-q, C-s, D-p

10. The parabola $y^2 = 4ax$ has a chord AB joining points $A(at_1^2, 2at_1), B(at_2^2, 2at_2)$.

	Coloumn - I	Coloumn - II
(A)	If normal chord AB subtend 90° at origin, then	(p) $t_2 = -t_1 + 2$
(B)	If AB is a focal chord then	(q) $t_2 = \frac{-4}{t_1}$
(C)	If AB subtend 90° at point $(0,0)$ then	(r) $t_2 = \frac{-1}{t_1}$
(D)	If AB if inclined at 45° to axis of parabola then	(s) $t_2 = -t_1 - \frac{2}{t_1}$

- 1) A-s, B-p, C-r, D-s 2) A-p, B-s, C-q, D-r
3) A-q, B-r, C-s, D-p 4) A-p, B-r, C-p, D-s

11.	Coloumn - I	Coloumn - II
(A)	Parabola $y^2 = 4x$ and the circle having its centre at $(6,5)$ intersects at right angle at the point (a,a) then one value of a is equal to the angle between the tangent drawn to $(y-2)^2 = 4(x+3)$ at the points where it is	(p) 13
(B)	intersected by the line $3x - y + 8 = 0$ is $4p/p$, then p has the value equal to	(q) 8
(C)	If the line $x - 1 = 0$ is the directrix of the parabola $y^2 - kx + 8 = 0$ then one of the value of k is / are	(r) $10\sqrt{5}$
(D)	Length of the normal of the parabola $y^2 = 8x$ at the point where abscissa & ordinate are equal is	(s) 4

- 1) A-p,B-q,C-s,D-r 2) A-s,B-q,C-p,D-r
 3) A-s,B-q,C-s,D-r 4) A-q,B-p,C-r,D-s

LEVEL - IV - KEY

- 1) 1 2) 1 3) 1 4) 1 5) 1 6) 3
 7) 1 8) 4 9) 1 10) 2 11) 3

LEVEL - IV - HINTS

- Both statements are correct
- The line joining the points passes through focus.
- $\cos^2 \alpha + \sin^2 \alpha - 2 \leq 0 \Rightarrow \sin^2 \alpha - \sin \alpha + 1 \geq 0$
- $x = 1$ is the axis of parabola.
- The minimum length of focal chord is latus rectum.
- If t_1 and t_2 are the parameters of A and B then

$$t_1 t_2 = -4 \text{ also } t_1 \left(-t_1 - \frac{2}{t_1} \right) = -4$$
- Clearly statement-1 is true. Also, the point of contact is given by $(a, -2a)$ & $(-2a, -a)$.
- Let the equation of the chord be $y = mx - 4am$. Clearly by homogenisation we can see that it subtends a right angle at the vertex. Now normal at $(8,8)$ does not satisfy $(8,0)$

- (A) Equation of the circle is $(x-r-1)^2 + y^2 = r^2$
 putting $y^2 = 4x$ and then $D = 0 \Rightarrow r = 4$
 (B) AB is focal chord $\text{Min AB} = \text{latus rectum} = 4a = 16$
 (C) Shortest distance along common normal
 Slope of common normal $= 0, \pm 2$
 So feet of normals will be $(4,4)$ & $(5,2)$
 or $(4,-4)$ and $(5,-2) \therefore d = \sqrt{1+4} = \sqrt{5}$
 (D) Harmonic mean $= 2a = 4$
- (A) Equation of the normal $y + t_1 x = 2at_1 + at_1^3$
 Slope of the normal $= -t_1$
 Slope of the normal $= \frac{2a(t_2 - t_1)}{a(t_2^2 - t_1^2)} = \frac{2}{t_2 + t_1}$
 $\Rightarrow -t_1 = \frac{2}{t_1 + t_2} \Rightarrow -t_1^2 - t_1 t_2 = 2$
 $\Rightarrow -t_1 = \frac{2}{t_1} \Rightarrow t_2 = -t_1 - \frac{2}{t_1}$
 (B) $t_1 t_2 = -1 \Rightarrow t_2 = -\frac{1}{t_1}$
 (C) $t_1 t_2 = -4 \Rightarrow t_2 = -\frac{4}{t_1}$
 (D) Slope of $AB = \tan 45^\circ = 1$
 Again slope $= \frac{2}{t_1 + t_2} = 1 \Rightarrow t_1 + t_2 = 2, t_2 = -t_1 + 2$
- (A) Point (a,a) lies on $y^2 = 4x \therefore a^2 = 4a$ i.e.,
 $a = 0, 4$
 (B) The line $3x - y + 8 = 0$ passes through the focus $(-2,2)$ so the tangents at the end points on the chord intersect at right angles $\Rightarrow p = 8$
 (C) $y^2 = k(x - 8/k)$ equation of directrix is
 $x - \frac{8}{k} = -\frac{k}{4} \Rightarrow x = \frac{8}{k} - \frac{k}{4}$
 compare with $x = 1 \Rightarrow \frac{8}{k} - \frac{k}{4} = 1 \Rightarrow k = 4$
 (D) end points of the normal chord will be $(8,8)$
 and $(2(-3)^2, 2 \cdot 2 \cdot (-3))$
 $\therefore$ length of the chord will be $10\sqrt{5}$.
